

Fish immunity

目錄

- Fish stress
- Fish immunity
- Fish immune response to bacteria
- Fish immune response to the virus
- Vaccine introduction



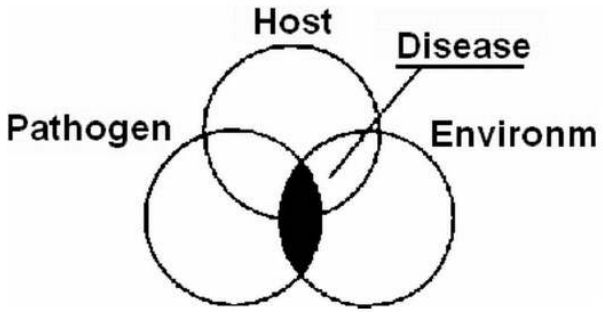
Aim

了解魚類免疫以利後續疫苗及魚類診斷的相關知識



Disease Resistance

- All fish do not get sick and die each time a disease outbreak occurs. There are many factors that affect how an individual responds to a potential pathogen. The pathogen (bacteria, parasite, or virus) must be capable of causing disease.



以下哪種生物可以引起魚類疾病？

A. 真菌

B. 細菌

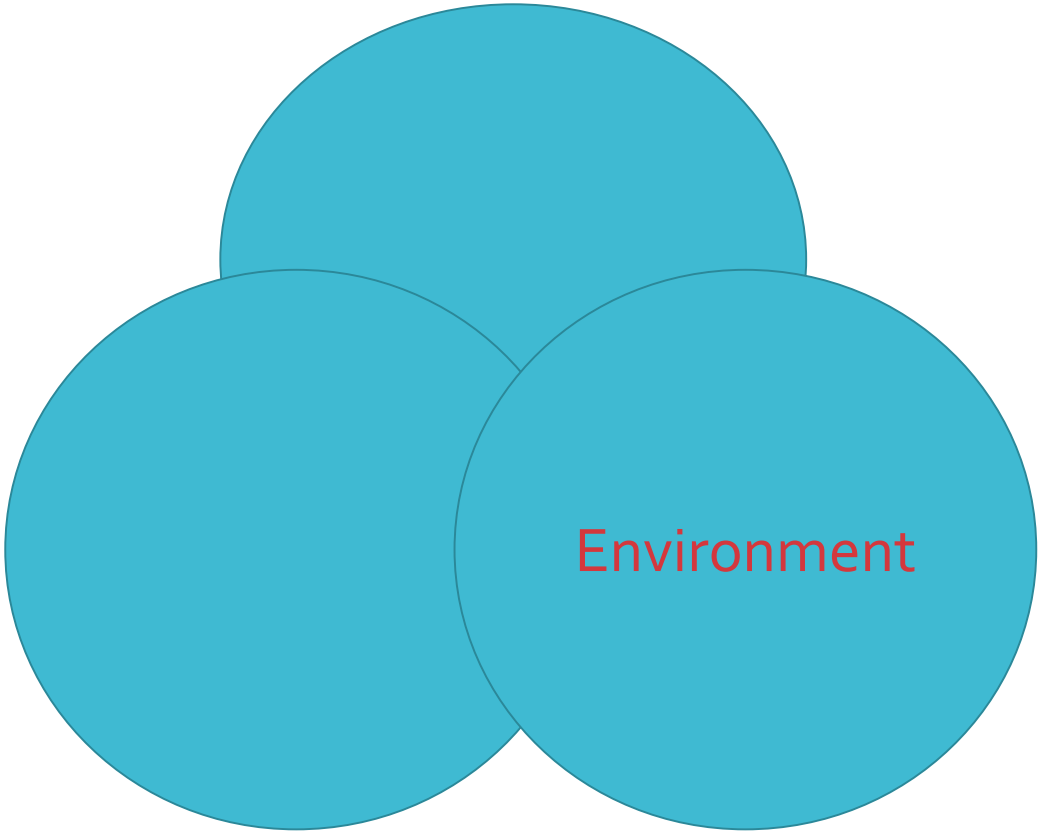
C. 病毒

D. 以上皆是

Answer

正確答案: D

Multiple Choice


Environment

Environment -stress

- 養殖環境的惡化會對養殖魚蝦造成緊迫，影響內分泌。環境變動以及緊迫反應中，為了調節生理恆定作用而產生改變的內分泌有皮質醇與兒茶酚胺。當魚類受緊迫與適應緊迫可分成三階段。
- **第一階段：反抗期（resistance）:**魚產生迴避行為，並藉生理調節，恢復原有之生理恆定性，主要是藉由中樞神經系統，將緊迫訊息匯集於下視丘前葉（preoptic anterior hypothalamus），一方面經由由自律神經傳遞訊息給嗜鉻細胞（chromaffin cells），促使其分泌兒茶酚胺釋放於血液中，產生應急反應，此時魚類產生抗拒與迴避行為，另一方面下視丘前葉分泌促腎上腺素釋放素（corticotropic releasing factor；CRF），CRF刺激腦下垂體（pituitary）分泌促腎上腺素（adrenocorticotropin hormone；ACTH）
- **第二階段：適應期（adaptation）:**在長期受到刺激下，體內生理調節逐漸達到新的生理恆定狀態。
- **第三階段：衰竭期（exhaustion）:**當刺激過強或持續過久，個體無法及時調整，或耗盡所儲存之能量與激素前驅物（hormone precursor），失去生理平衡，而導致死亡。



以下哪一項是魚類在第一階段反抗期時所產生的應對反應？

A. 增加進食

B. 產生迴避行為

C. 改變繁殖行為

D. 減少活動量

Answer

正確答案: B

Multiple Choice

Environment -stress

- 血液中皮質醇濃度增加會抑制吞噬細胞（phagocytes）
- 降低血液中循環性T樣淋巴球（circulating T-like lymphcytes）及循環性B樣細胞淋巴球（circulating B-like lymphcytes）。

- 在緊迫反應下淋巴球所產生的變化，目前還有很多尚未釐清，鱒魚Oncorhynchus Kisutch 受到急性緊迫（acute stress）或直接注射皮質醇，會使血液中循環性淋巴球趨向於胸線（thymus）及頭腎（head kidney）淋巴組織，造成血液中循環性淋巴球數量降低。於體外（in vitro）實驗中，發現糖皮質酮（glucocortioids），抑制骨髓（bone marrow）中單核球（monocytic）以及顆粒性球（granulopoietic）之前驅細胞的分化與增生。

血液中皮質醇濃度增加的影響是什麼？

A. 提高免疫系統的活動

B. 抑制吞噬細胞和降低淋巴球數量

C. 增加淋巴球的生成

D. 促進骨髓增生

Answer

正確答案: B

Multiple Choice

Stress

- The nonspecific response of the body to any demand made upon it .(Hans Selye (1973)



Stress

- The response to stress is considered an ***adaptive mechanism*** that allows the fish to cope with real or perceived stressors in order to maintain its normal or homeostatic state

- Quite simply, stress can be considered as a state of threatened homeostasis that is re-established by a complex suite of adaptive responses



在魚類反應中的壓力被認為是一種什麼樣的機制？

A. 生理機制

B. 適應機制

C. 行為機制

D. 防禦機制

Answer

正確答案: B

Multiple Choice

Stress

- Stress in fish is an habitual, undesirable aspect of production
- Stress results from biotic and/or abiotic challenges that act in changing or modifying the animal's natural or homeostatic state.
- Common fish stressors involve alterations in the fish's immediate environment
- **Chemical changes - contaminants, low oxygen and acidification**
- **Physical changes - handling, capture, confinement and transport**
- **Perceived changes (感官變化) - startling or predators**



Stress

- The above stressors are categorized as either acute or chronic depending on their duration and frequency
- ► *Acute fish stressors* are events which the animal experiences for a short period of time such as handling
- ► *Chronic fish stressors* are defined as a constant or recurring exposure that causes a prolonged physiological response
- Fish stressors of both types have severe negative effects on the animal's growth and health



Stress

- What Causes Stress?



Stress

1. Chemical stressors
2. Biological stressors
3. Physical stressors
4. Procedural stressors



Chemical stressors

1. Poor Water Quality
2. Pollution
3. Diet composition
4. Nitrogenous and other metabolic wastes



Biological stressors

1. Population density
2. Other species of fish
3. Microorganisms
4. Macroorganisms

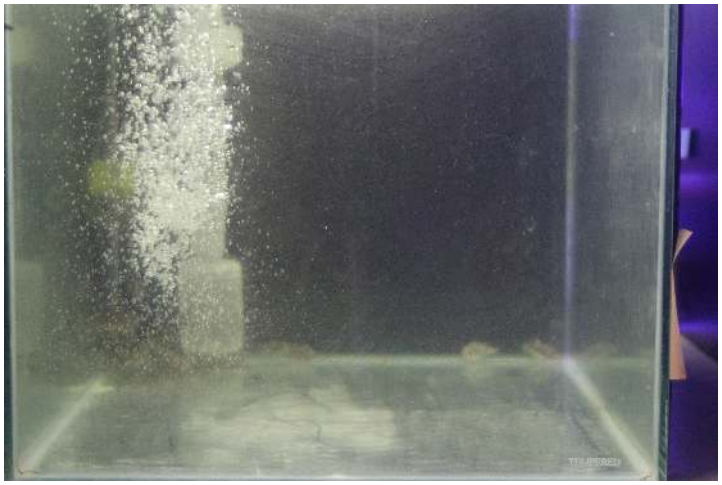


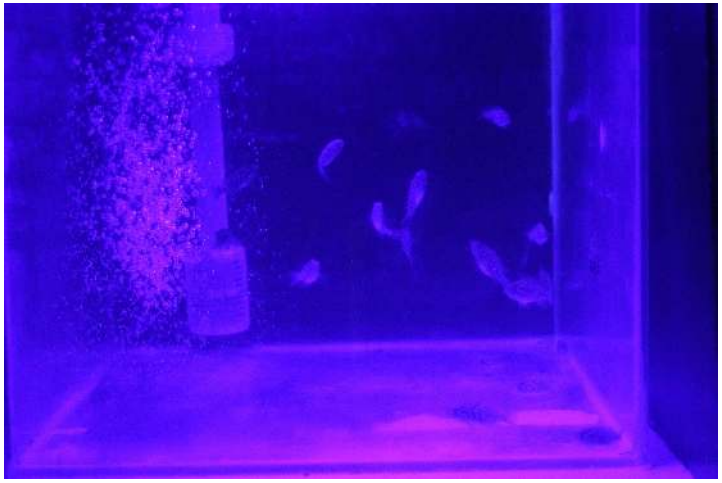
Physical stressors

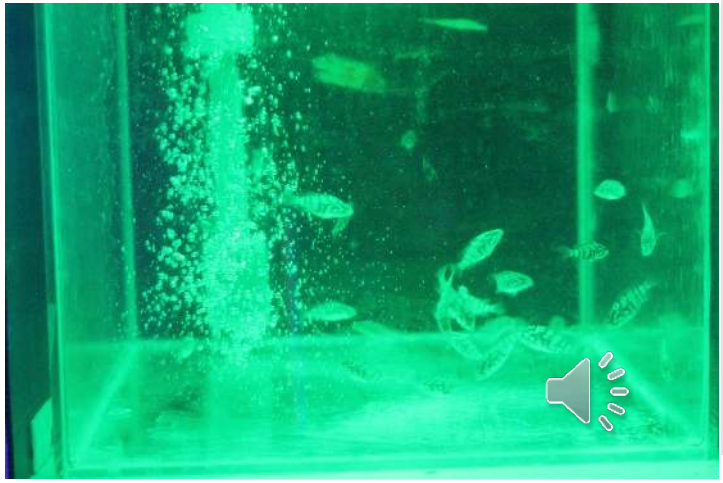
1. Temperature:
2. Light
3. Sounds/noise



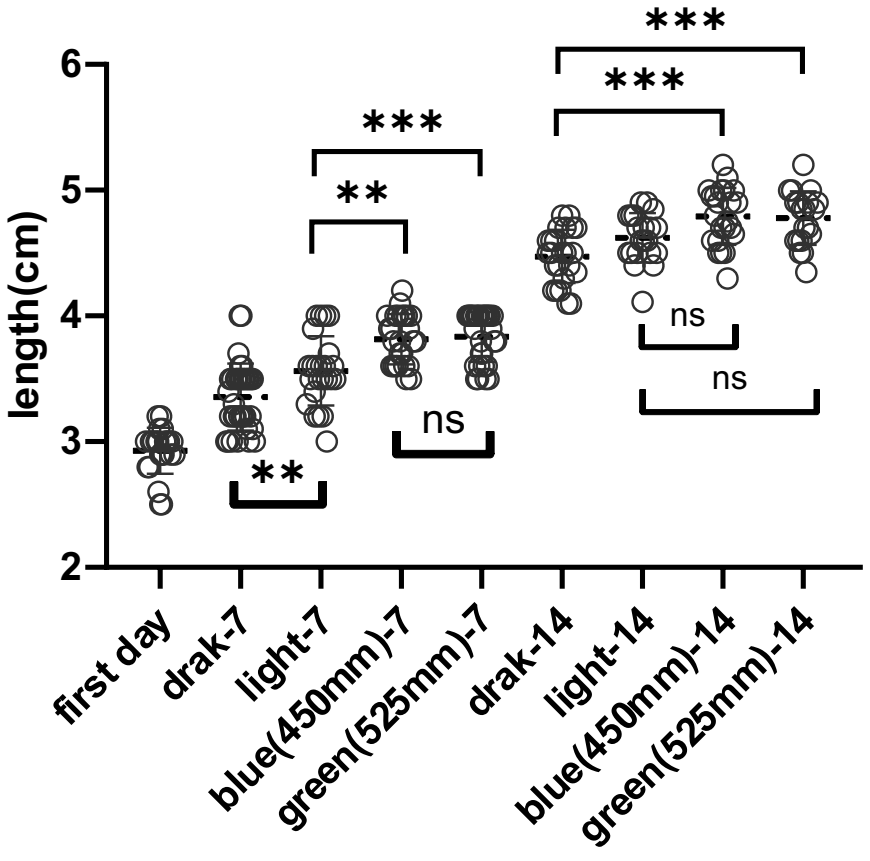
Light



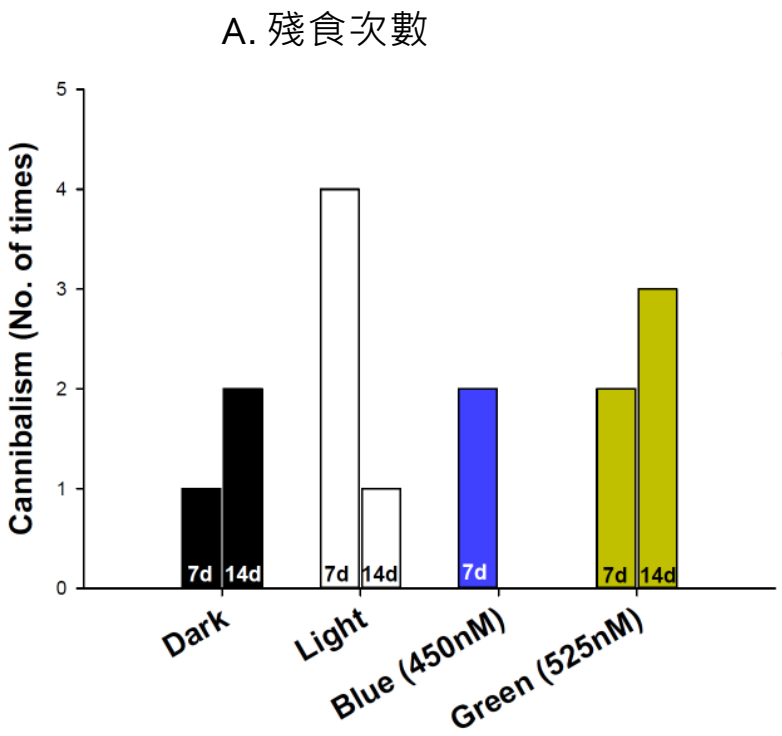


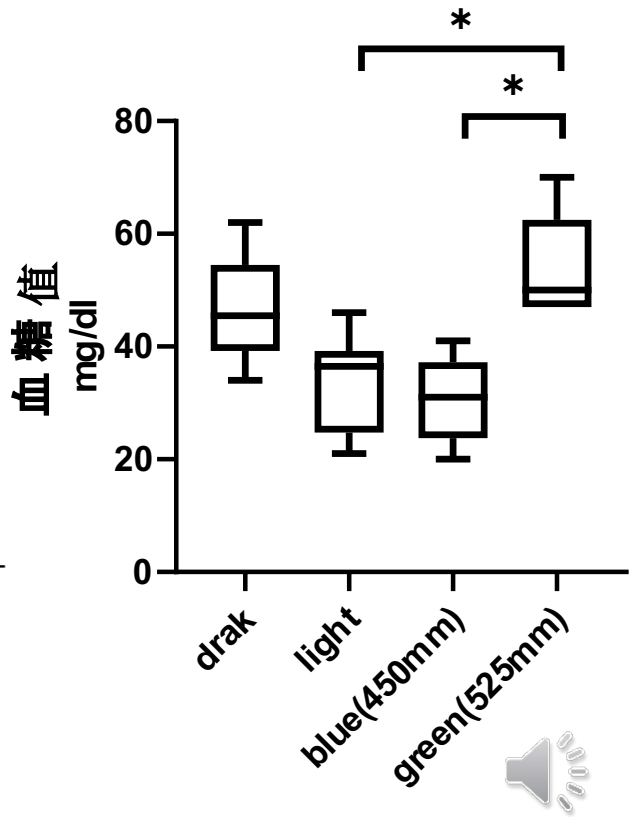


Light

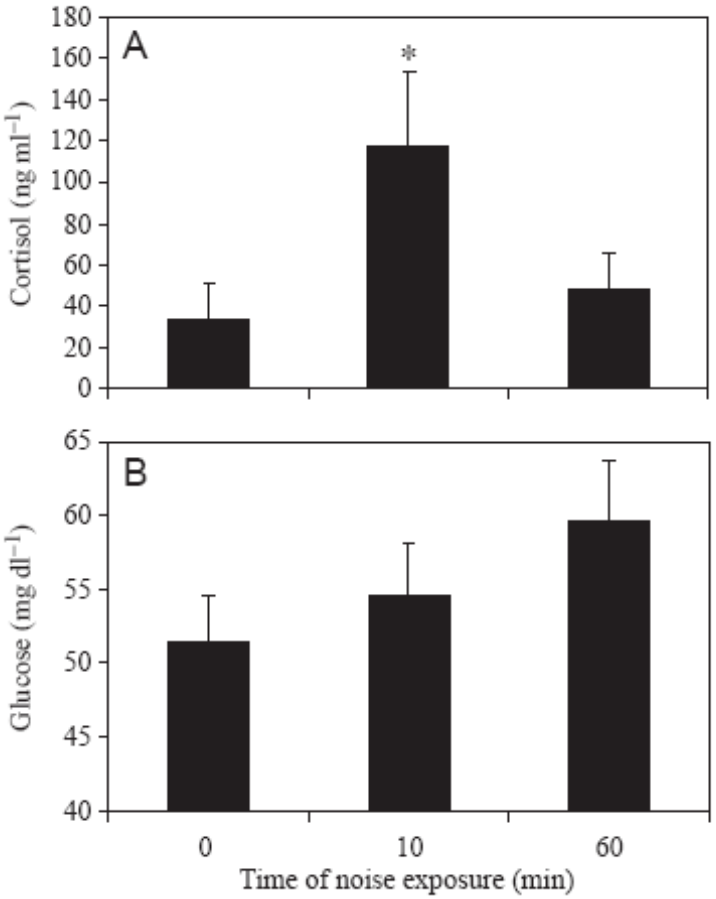

length(cm)
6
5
4
3
2
***
***
***
**
**
ns
ns
ns
first day
drak-7
light-7
blue(450mm)-7
green(525mm)-7
drak-14
light-14
blue(450mm)-14
green(525mm)-14

Light


A. 殘食次數
Cannibalism (No. of times)
0
1
2
3
4
5
7d
14d
7d
14d
7d
7d
14d
Dark
Light
Blue (450nM)
Green (525nM)


*
*
血糖值
mg/dl
0
20
40
60
80
drak
light
blue(450mm)
green(525mm)

Noise



Smith et al., 2004. Journal of Experimental Biology 207: 427-435

Effects of noise:
Induces a stress response in exposed fish

Can induce temporary hearing loss

Specialists more sensitive

Fish can recover over time

Procedural stressors

1. Handling
2. Shipping
3. Disease Treatments



Behavior in Fish after stress

- One of the immediate signs of a stressed state in fishes is a behavioral response.
- **Food acquisition**
- **Predator avoidance**
- **Prey capture**
- **Migration**
- **Habitat preference**

Indicators of environmental stressors



The Physiological Response to Stress in Fish

- Generally, stress in fish is a non-specific response classified as three distinct responses:

- **Primary response**
- The primary response is characterized by the activation and the secretion of the hormones, ***catecholamines*** and, ***corticosteroids (cortisol)*** into the blood.



location

- Chromaffin tissue (near kidneys and heart): secrete catecholamine

(epinephrine/norepinephrine) - **'fight or flight'** response

- Interrenal: secretes steroids (cortisol, corticosterone, cortisone)

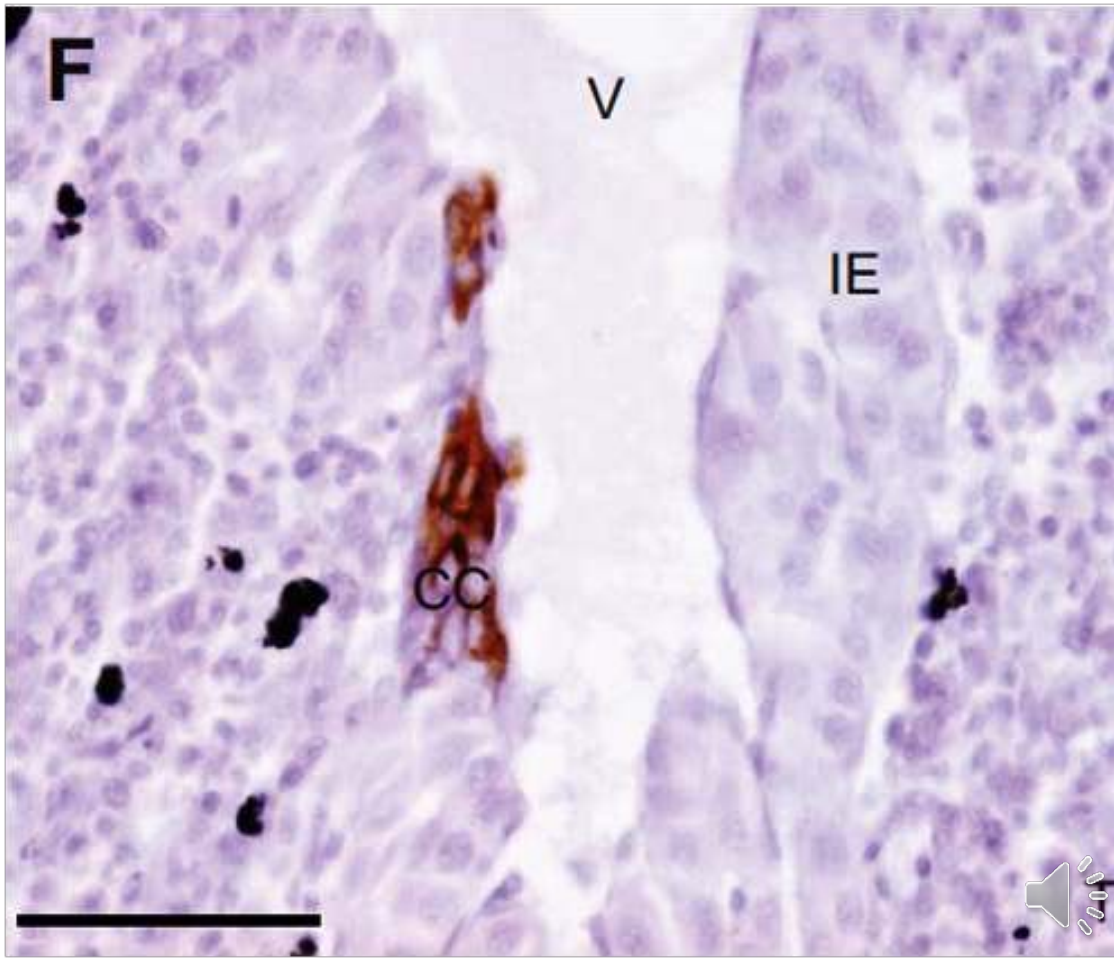
- **stress response, osmoregulation, protein/carbohydrate metabolism**



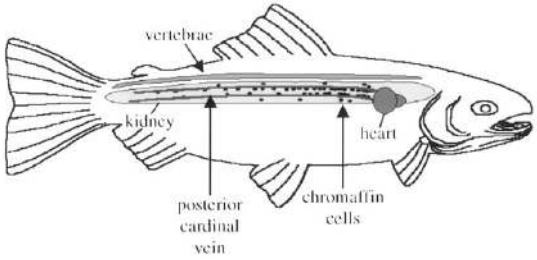
location

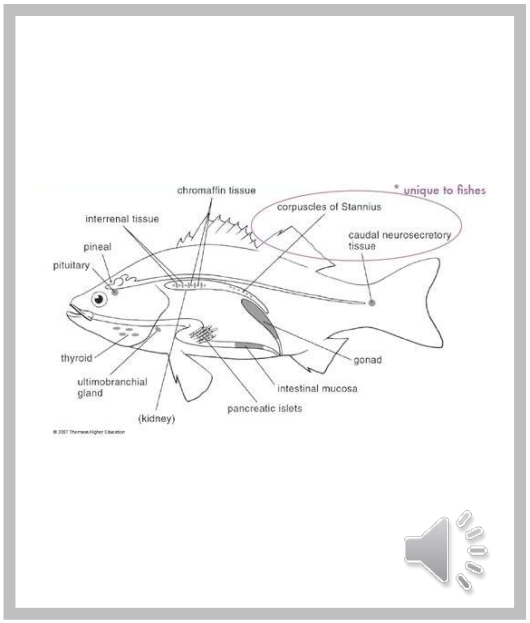
The **chromaffin cells** are identified by the positive reaction for tyrosine hydroxylase (**TH**).

CC, chromaffin cell; **IE**, interrenal tissues; **M**, melanocyte; **RT**, renal tissue; **V**, blood vessel



location


vertebrae
kidney
heart
posterior cardinal vein
chromaffin cells


chromaffin tissue
* unique to fishes
corpuscles of Stannius
interrenal tissue
caudal neurosecretory tissue
pineal
pituitary
thyroid
gonad
ultimobranchial gland
intestinal mucosa
pancreatic islets
(kidney)

The Physiological Response to Stress in Fish

- Generally, stress in fish is a non-specific response classified as three distinct responses:

- **Secondary response 2**
- The release of hormones triggers the secondary response which involves *the release of glucose* into the blood for energy production, followed by increases in heart rate, gill blood flow and metabolic rate which causes changes to blood lactate and hematocrit (reduction in circulating lymphocytes)



以下哪一項不是魚類在應對壓力時的生理變化？

A. 心率增加

B. 血乳酸增加

C. 白血球數量增加

D. 鰓血流量增加

Answer

正確答案: C

Multiple Choice

The Physiological Response to Stress in Fish

- Stress is an energy-demanding process, and the animal has to mobilize energy substrates to metabolically cope with stress
- One of the common indicators of metabolic effects due to stress is the increase in plasma glucose concentration
- The production of glucose with occurrence of stress assists the animal by providing energy substrates to tissues such as the brain, gills, and muscles to cope with the increased energy demand
- Adrenaline (short-term) and cortisol (long-term) also increase glucose production in fish



The Physiological Response to Stress in Fish

- **Tertiary response 3**
- Changes in blood physiology ultimately cascade into a whole body change or the tertiary stress responses. Changes associated with the tertiary stress response include **reduced growth rate**, **decreased disease resistance**, **altered behavior** and **reduced survivability**.



魚類在面對壓力時，第三級反應會導致哪些身體變化？

A. 增加生長速率

B. 提高疾病抵抗力

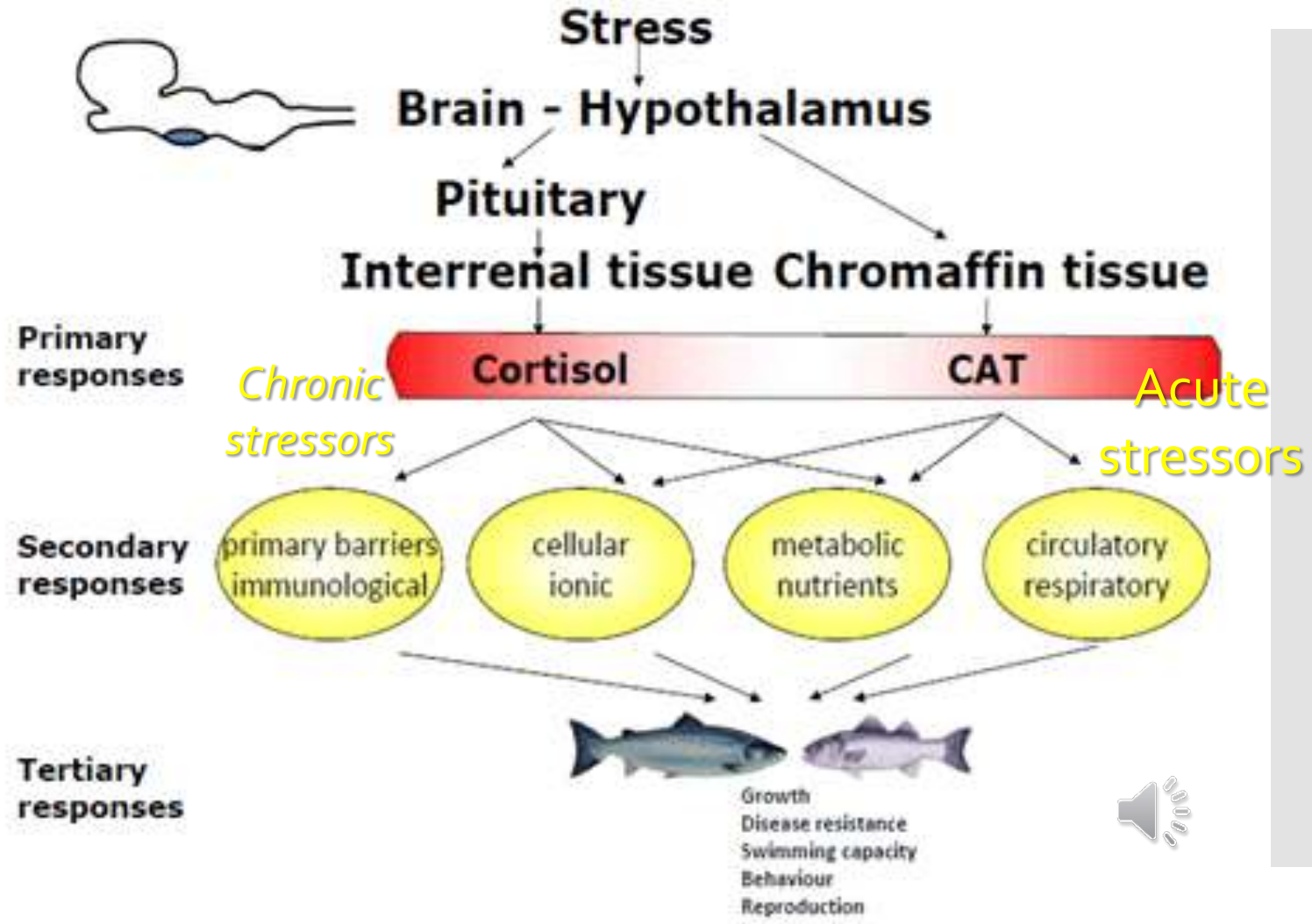
C. 行為改變

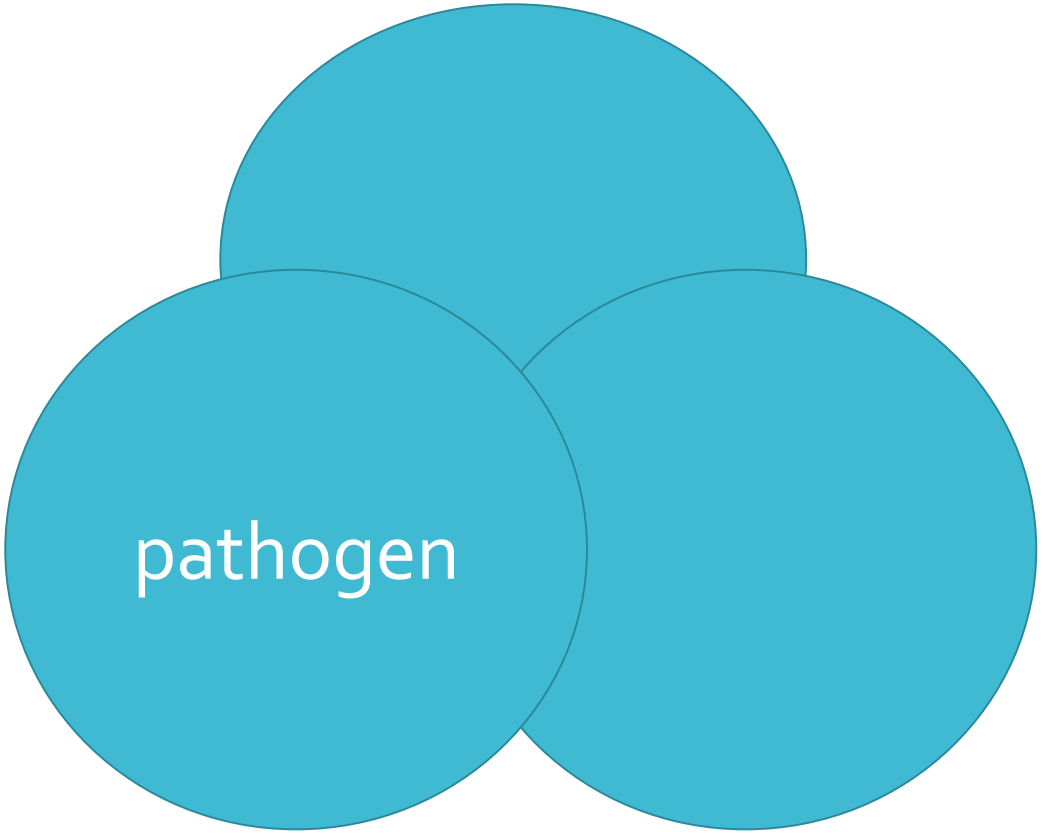
D. 改善存活率

Answer

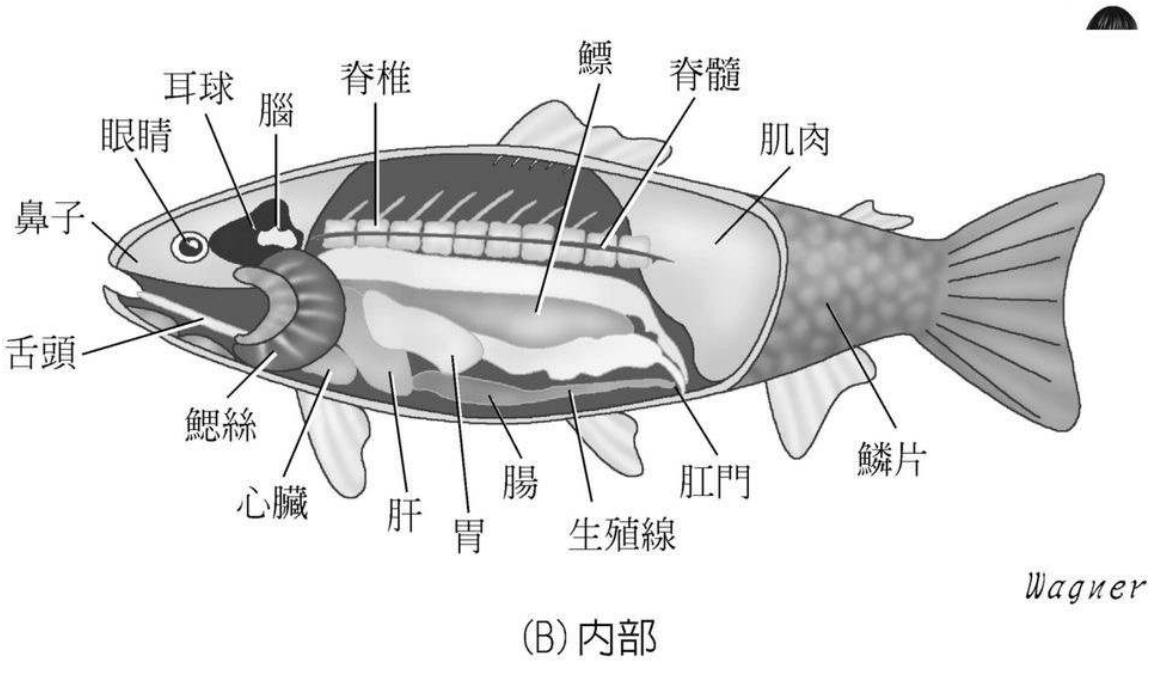
正確答案: C

Multiple Choice


Stress
Brain - Hypothalamus
Pituitary
Interrenal tissue
Chromaffin tissue
Primary responses
Cortisol
CAT
Chronic stressors
Acute stressors
Secondary responses
primary barriers immunological
cellular ionic
metabolic nutrients
circulatory respiratory
Tertiary responses
Growth
Disease resistance
Swimming capacity
Behaviour
Reproduction


pathogen

魚類最先接觸病原的位置

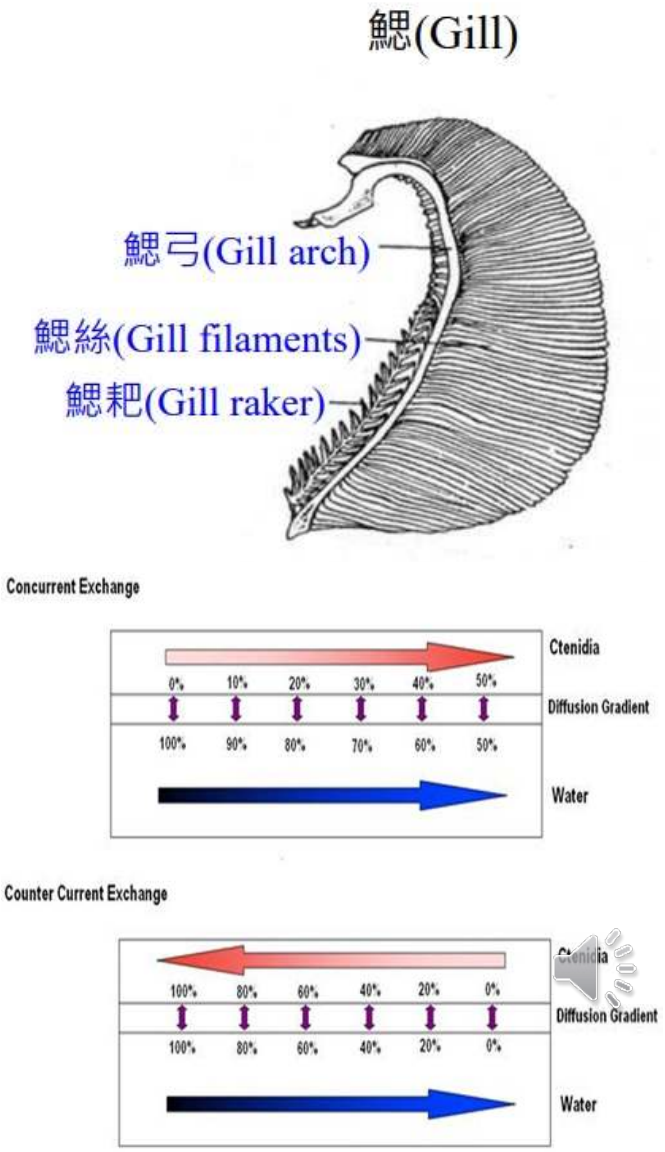


(B) 內部

圖 6-1　魚類的器官（續）

GILL

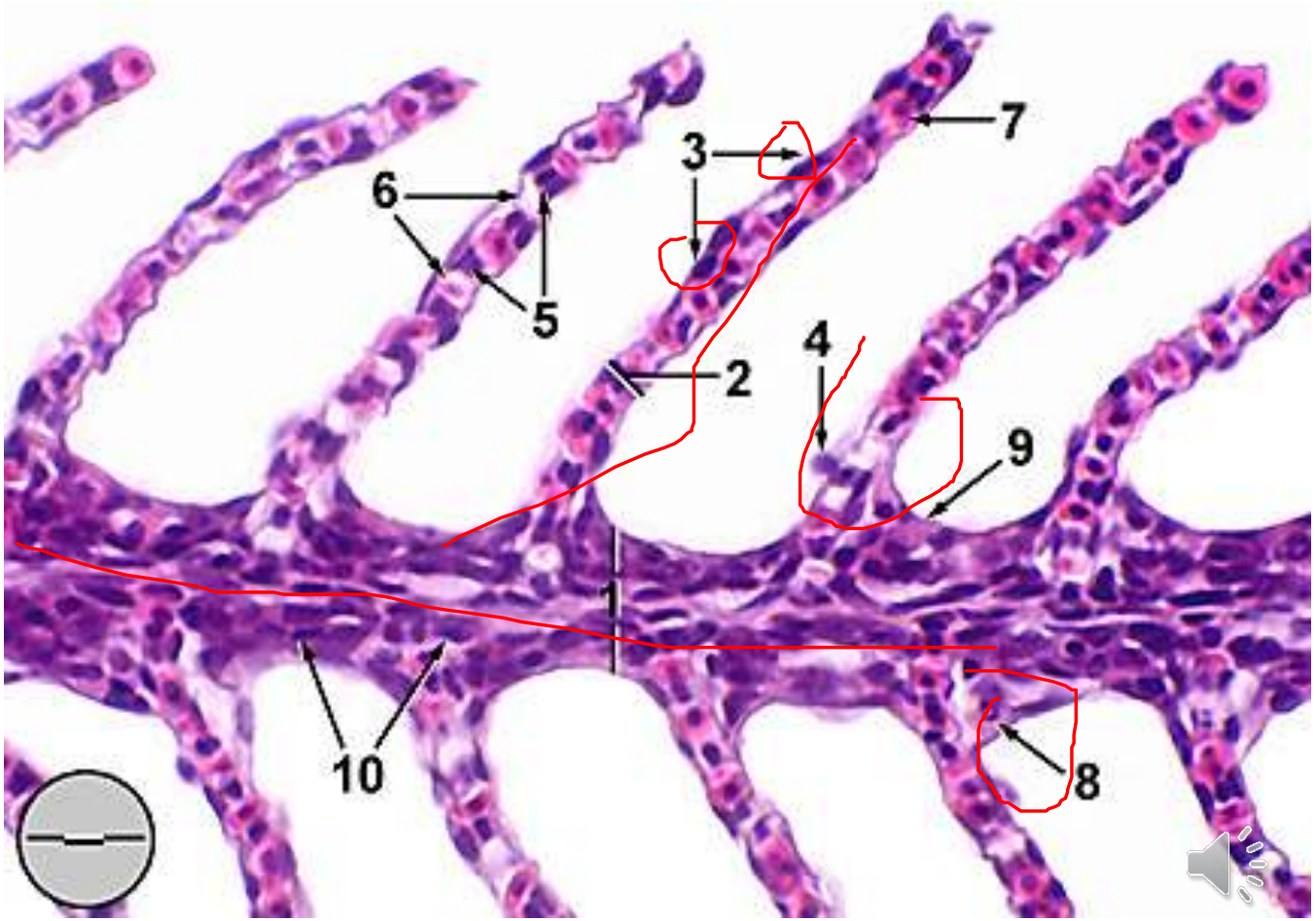
- Gas exchange
- Ionoregulation
- Acid-base regulation
- Detoxification– nitrogen waste



GILL

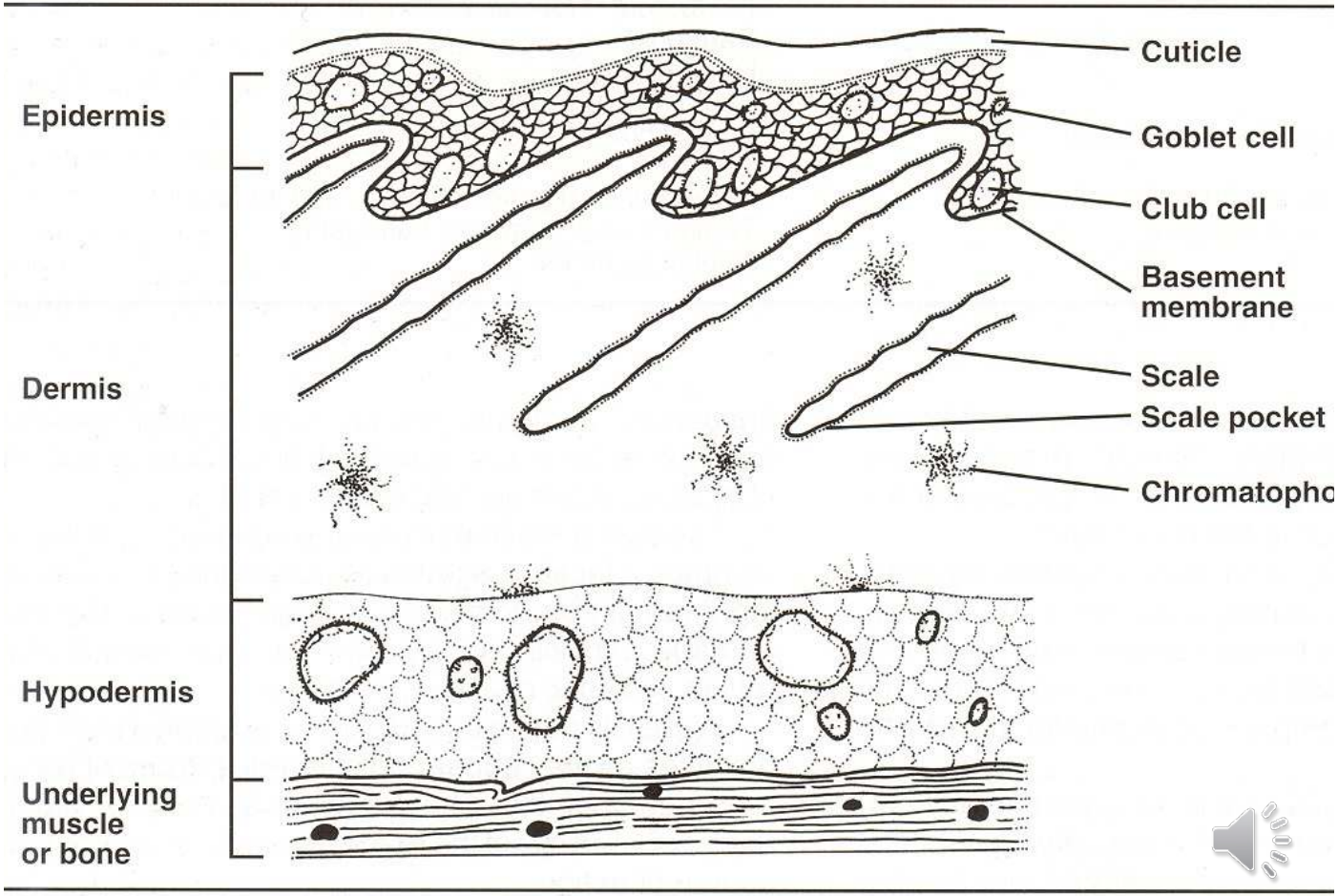
(Formalin, H&E, Bar = 16.7 µm).

1. primary lamella
2. secondary lamella
3. epithelial cell
4. mucous cell
5. pillar cell
6. lacuna (capillary lumen)
7. erythrocyte within capillary lumen
8. chloride cell
9. rodlet cell
10. undifferentiated basal cell.



Skin

The skin of the fish contains an epidermis lacking dead cornified cells and dermis of connective tissue in varying densities.



Mucus

Mucus which by entrapping micro organisms and continuously inhibits colonization of the integument.

The rate of mucus production in fish can be increased in the response to infection.

Mucus of the fish is toxic to certain micro organisms.



Mucus

Antimicrobial peptides

Lysozymes



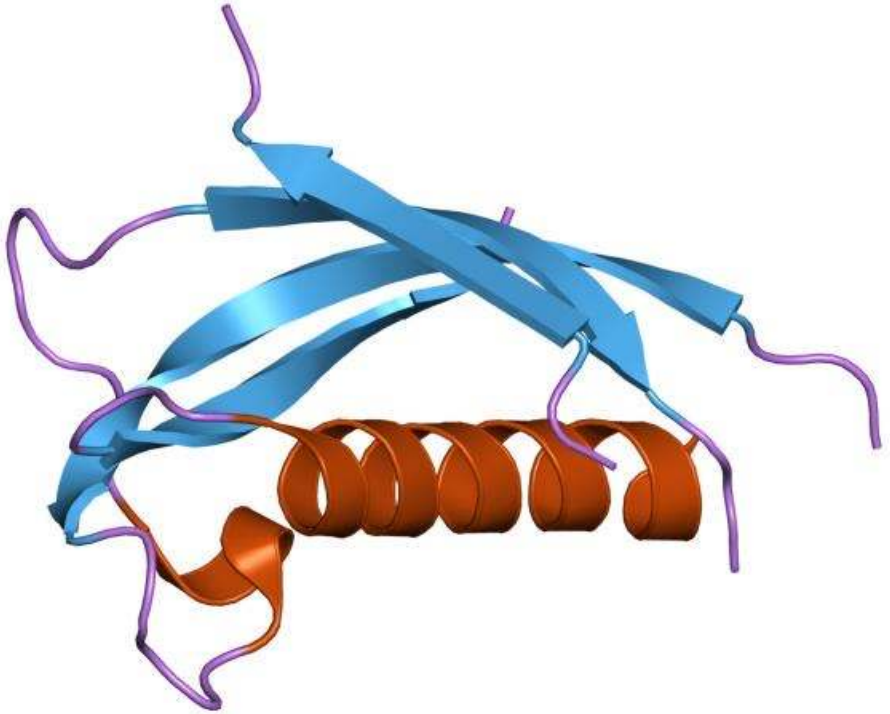
Mucous



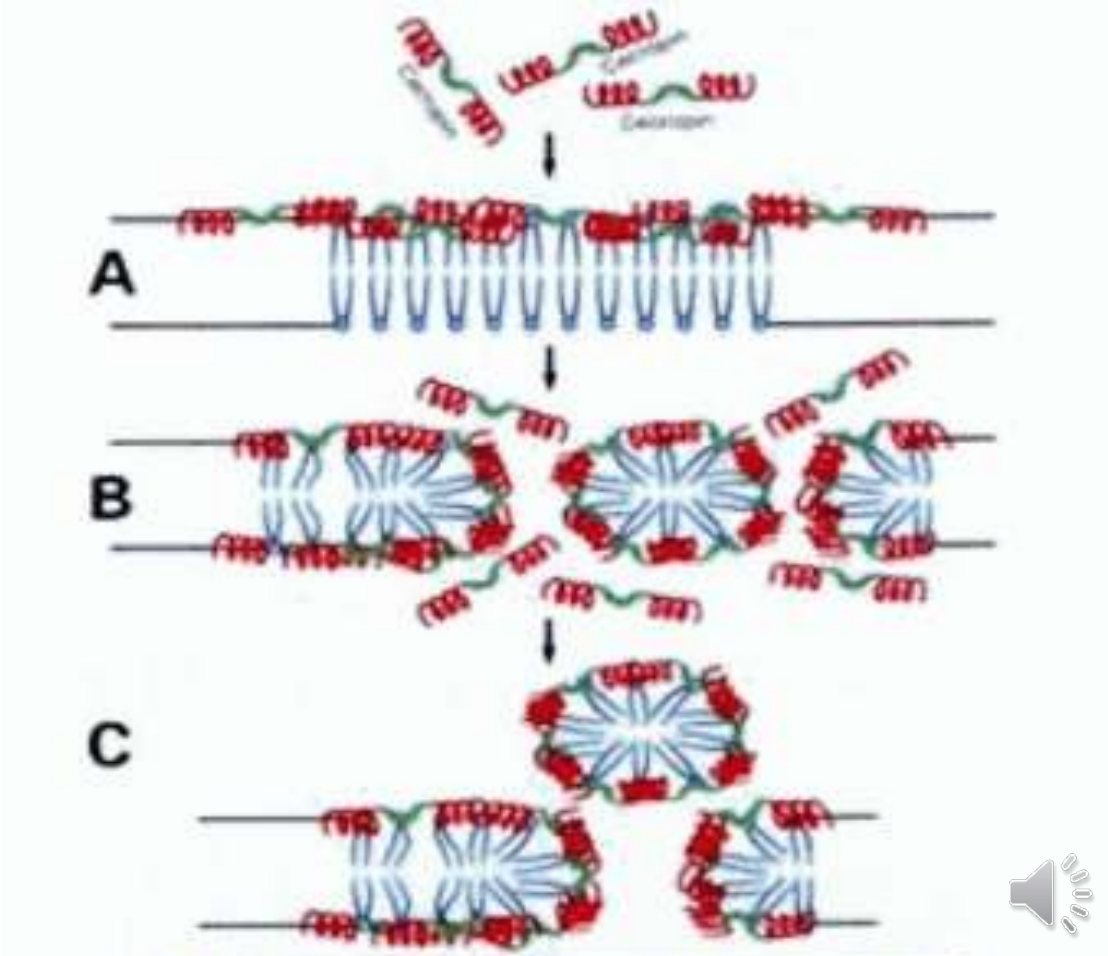


Antimicrobial peptides

- 12~80 amino acid
- cathelin domain



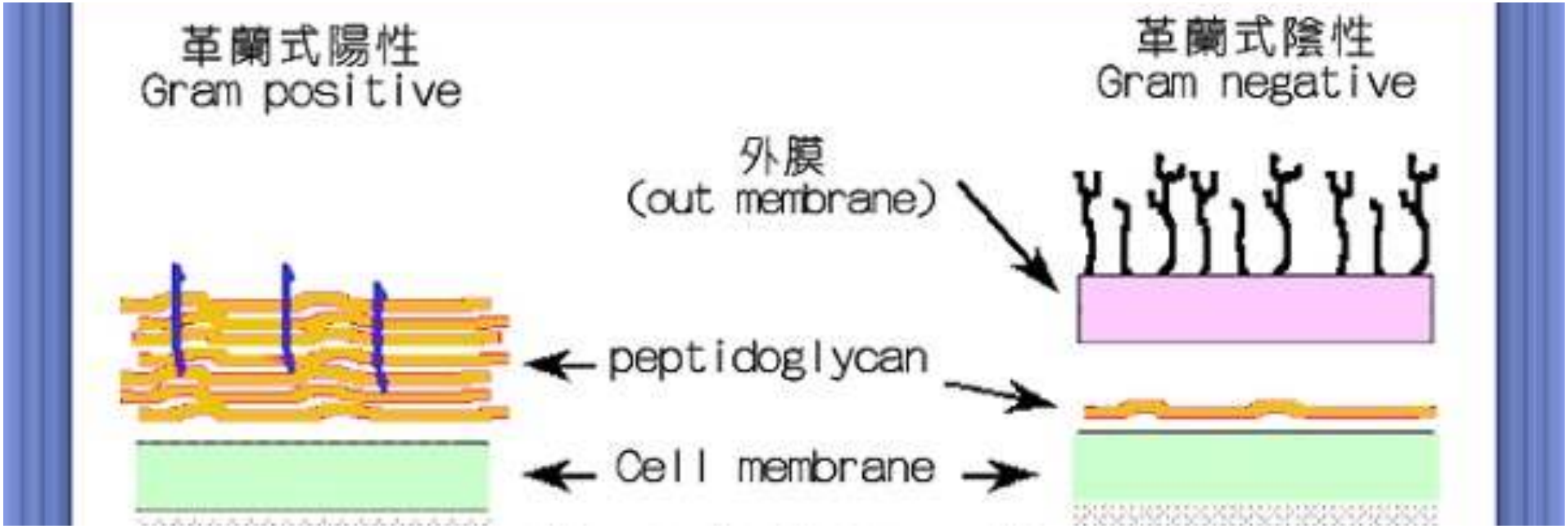
Antimicrobial peptides



Lysozymes

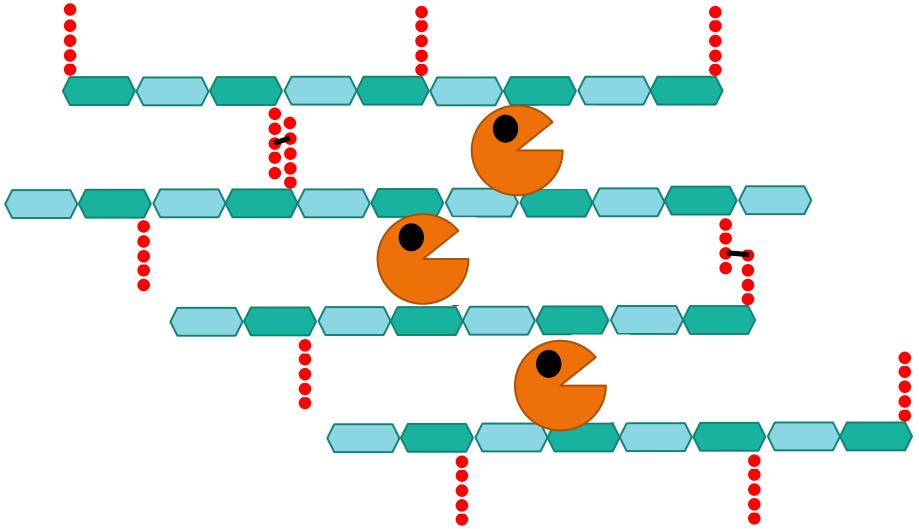
- Lysozyme is one of the most studied innate responses in fish.
- It can act on the peptidoglycan layer of the bacterial cell walls and resulting in the lysis of the bacteria.
- Lysozyme response in fish is very rapid not only related to bacteria, but also to other alarm situation such as after stress



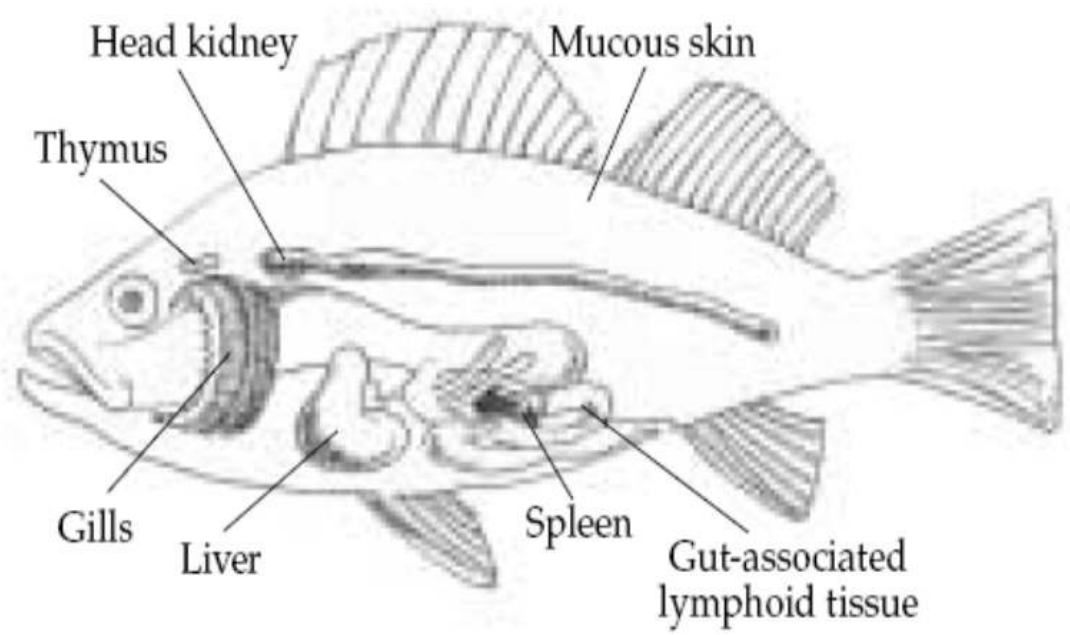

革蘭式陽性
Gram positive
革蘭式陰性
Gram negative
外膜
(out membrane)
peptidoglycan
Cell membrane

Lysozymes

Lysozymes

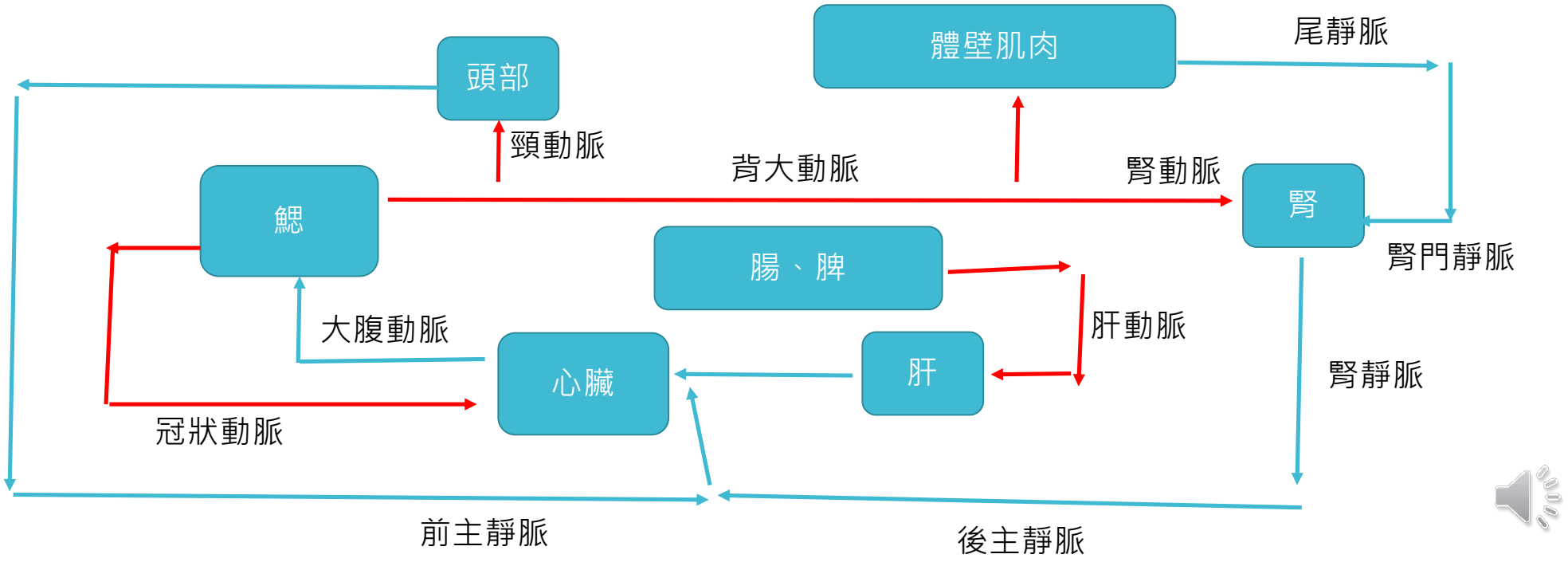


Lysozymes


Head kidney
Mucous skin
Thymus
Gills
Liver
Spleen
Gut-associated
lymphoid tissue

魚體內接觸病原的位置

Circulatory system



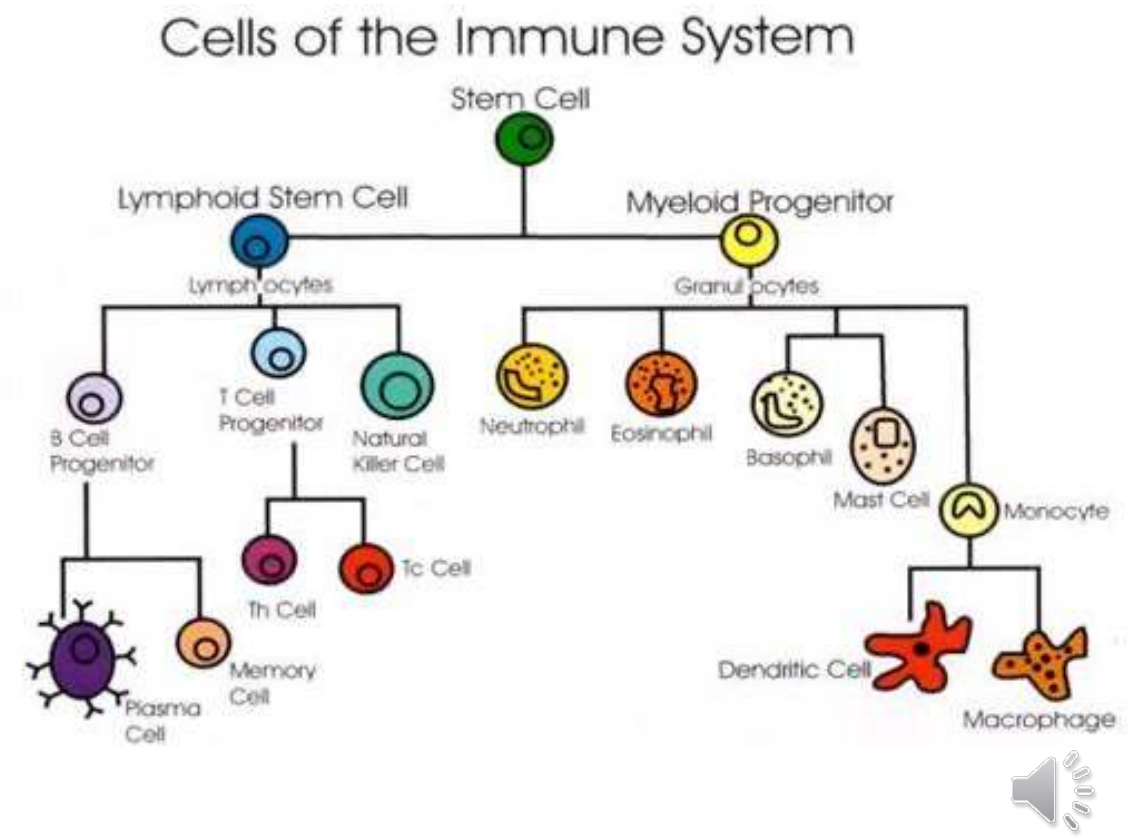
Immune Organs

- Kidney: blood flows slowly through kidney and antigens are "trapped" or exposed to reticular cells, macrophages, lymphocytes Anterior is where "memory" occurs
- Spleen: secondary to kidney, involved in immune reactivity and blood cell formation, contains lymphocytes and macrophages
- Liver: could be involved in production of components of the complement cascade, important in resistance; not real clear



Immune cell

Teleostean hemopoietic (blood-forming) sites are primarily the **kidney** and spleen.



Immune cell

- Reticuloendothelial system
- Comprise the highly **phagocytic cells** distributed in various **tissues** throughout the body



Non-specific immune cells

- **Monocytes and tissue macrophages**: most important cells in immune response, produce cytokines, primary cells involved in phagocytosis and first killing of pathogens upon first recognition and subsequent infection
- **Neutrophils**: primary cells in early stages of inflammation, neutrophils produce cytokines to recruit immune cells to damaged or infected area; neutrophils are phagocytic in I. Punctatus, kill bacteria by extracellular mechanisms
- **Natural killer like cells**: use receptor binding to target cells and lyse them; important in parasitic and viral immunity

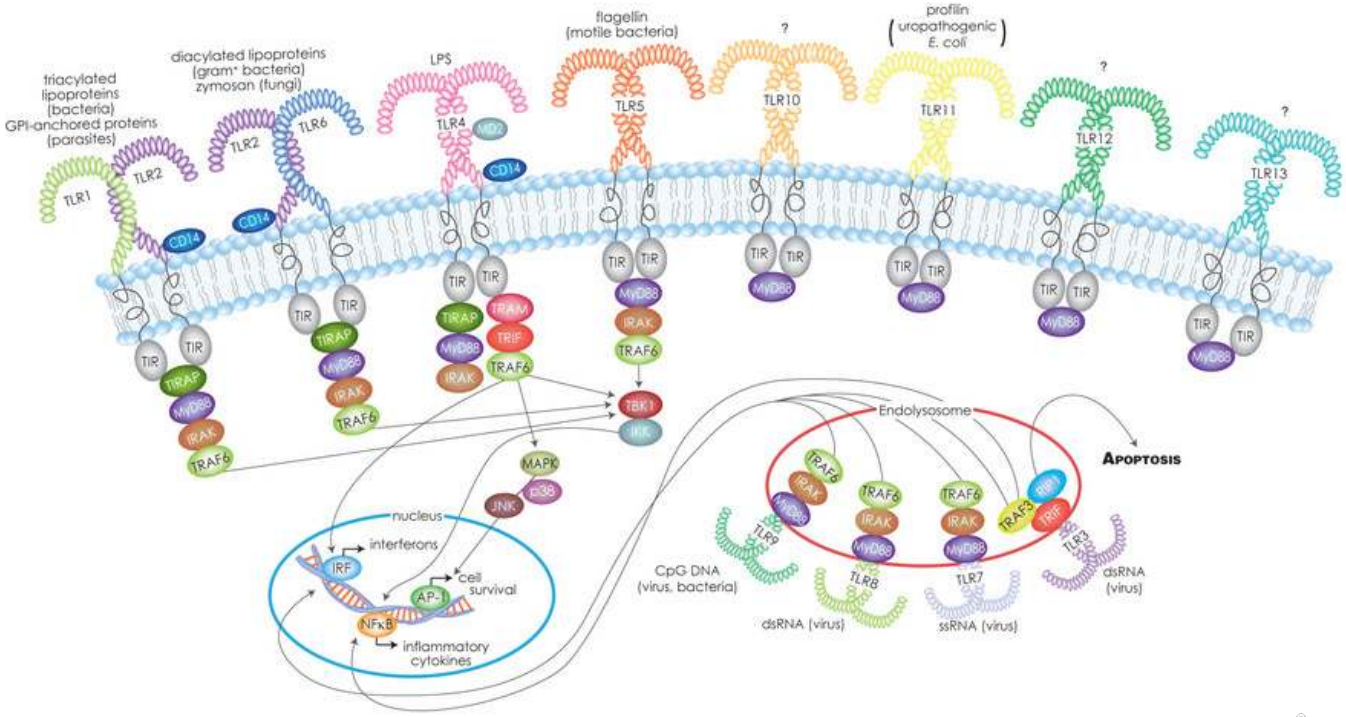


Pattern Recognition Receptors (PRR)

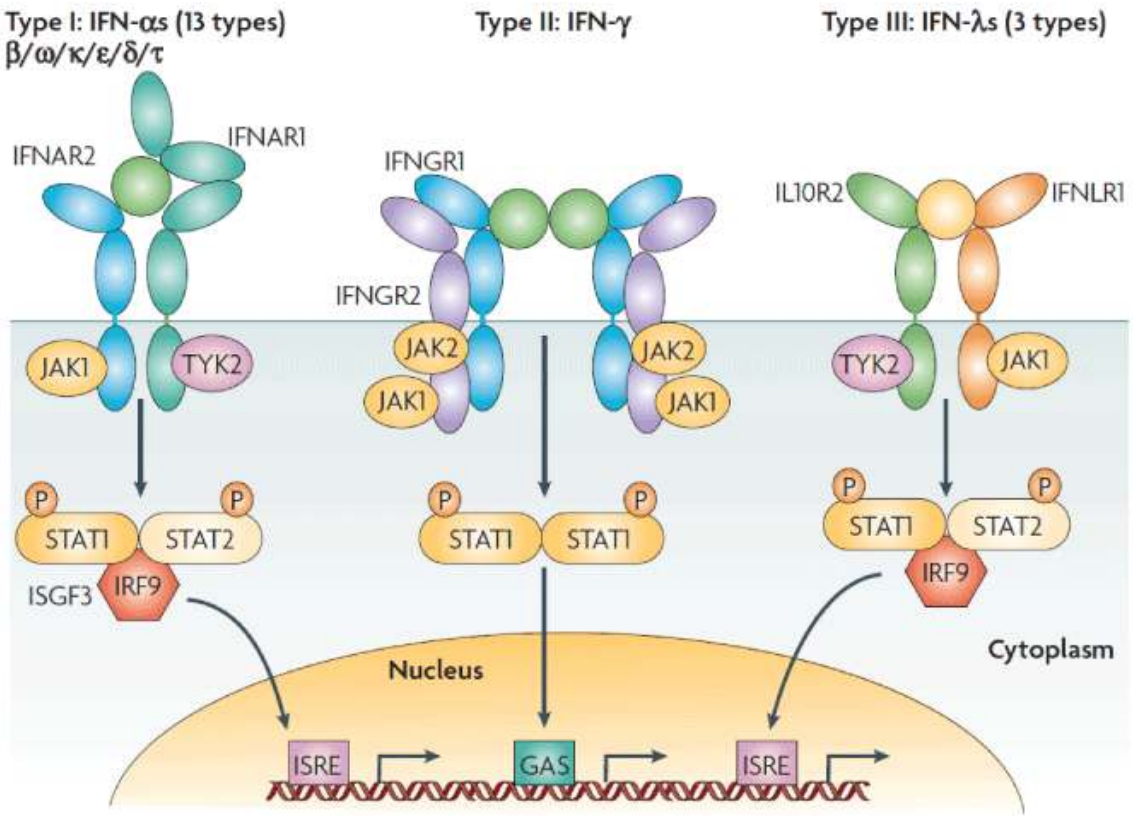
- Toll-like receptor(TLR)
  - Pathogen Associated Molecular Patterns (PAMP)
- CLR(C-type lectin receptor)
  - CHO on pathogen surface
- RLR (Retinoicacid-inducible gene-I-like receptors)
  - viral RNA in the cytosol of infected cells
- Nod-like receptor(NLR)
  - Pathogen Associated Molecular Patterns (PAMP)
  - Damage/Danger-associated molecular pattern (DAMP)
- Scavenger receptors
  - Damage/Danger-associated molecular pattern (DAMP)



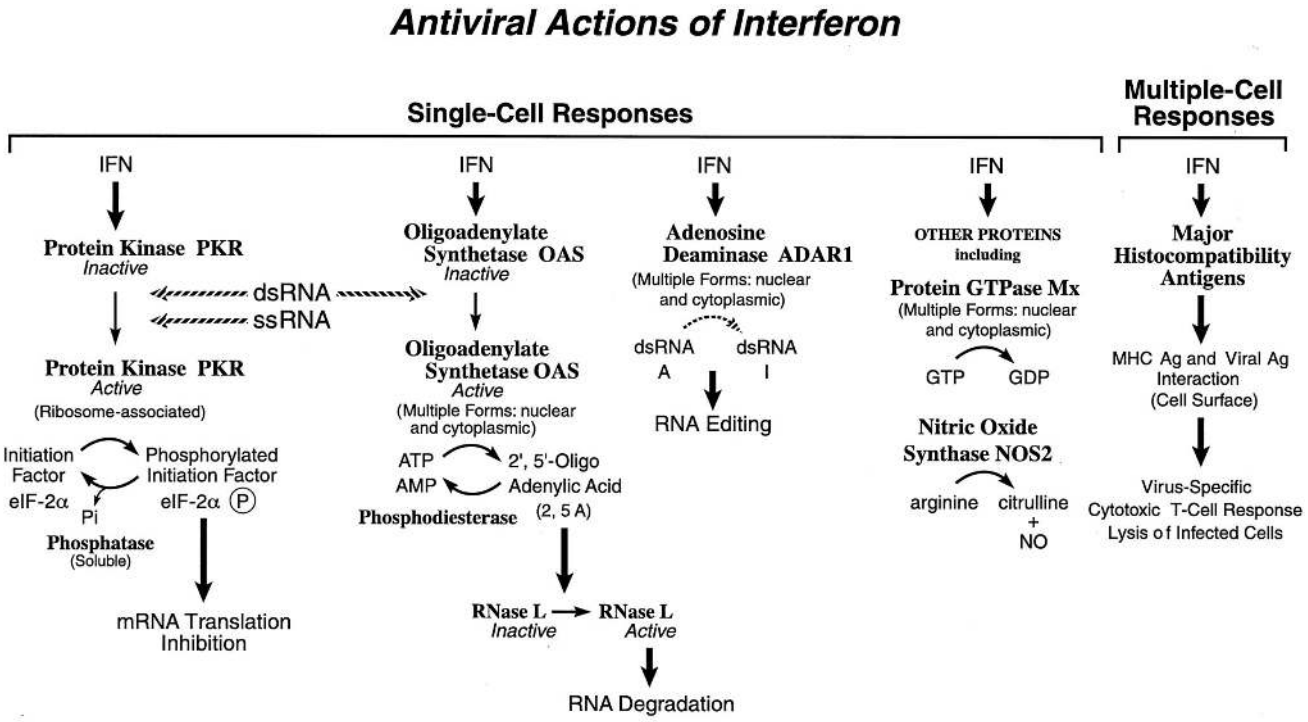
Toll-like receptor(TLR)



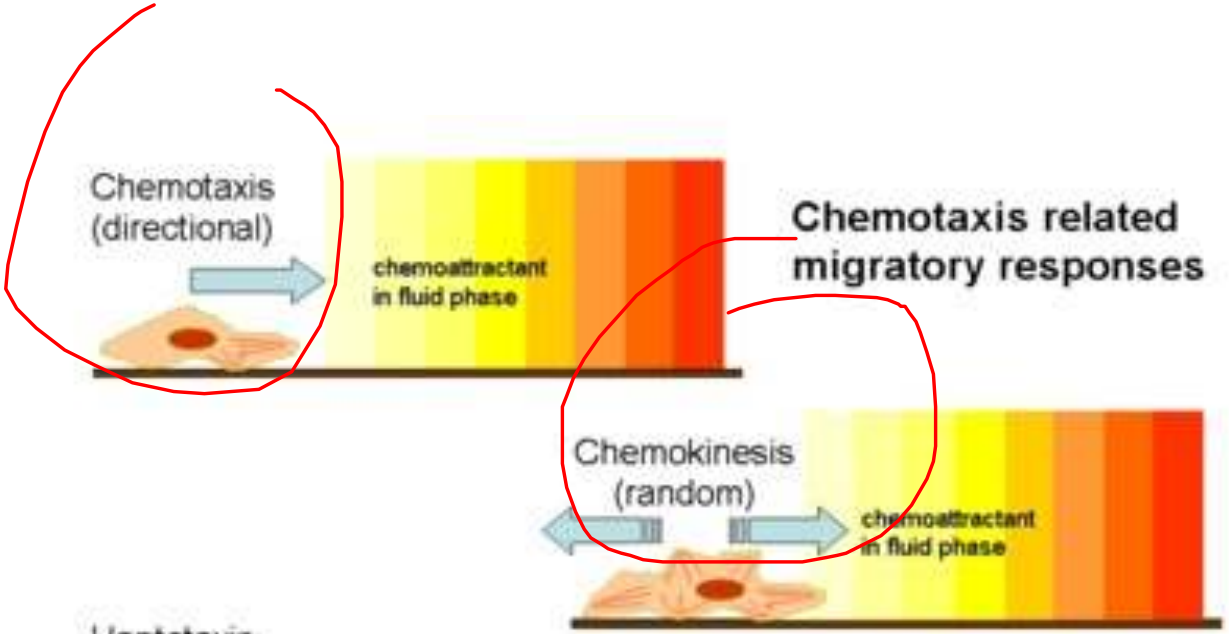
Toll-like receptor(TLR)



Interferon


Antiviral Actions of Interferon
Single-Cell Responses
Multiple-Cell Responses
IFN
Protein Kinase PKR
Inactive
dsRNA
ssRNA
Protein Kinase PKR
Active
(Ribosome-associated)
Initiation Factor eIF-2α
Pi
Phosphorylated Initiation Factor eIF-2α (P)
Phosphatase
(Soluble)
mRNA Translation Inhibition
IFN
Oligoadenylate Synthetase OAS
Inactive
Oligoadenylate Synthetase OAS
Active
(Multiple Forms: nuclear and cytoplasmic)
ATP
AMP
2', 5'-Oligo Adenylic Acid (2, 5 A)
Phosphodiesterase
RNase L Inactive
RNase L Active
RNA Degradation
IFN
Adenosine Deaminase ADAR1
(Multiple Forms: nuclear and cytoplasmic)
dsRNA A
dsRNA I
RNA Editing
IFN
OTHER PROTEINS including
Protein GTPase Mx
(Multiple Forms: nuclear and cytoplasmic)
GTP
GDP
Nitric Oxide Synthase NOS2
arginine
citrulline + NO
IFN
Major Histocompatibility Antigens
MHC Ag and Viral Ag Interaction (Cell Surface)
Virus-Specific Cytotoxic T-Cell Response Lysis of Infected Cells

Non-specific immune cells





Phagocytosis

- most primitive of defense mechanisms, occurs in stages
- Movement by chemotaxis (directional) or chemokinesis (non-d) of phagocytes in response to foreign object

1.Attachment via lectins

2.Engulfment of the foreign agent (simple movement into the phagocyte)

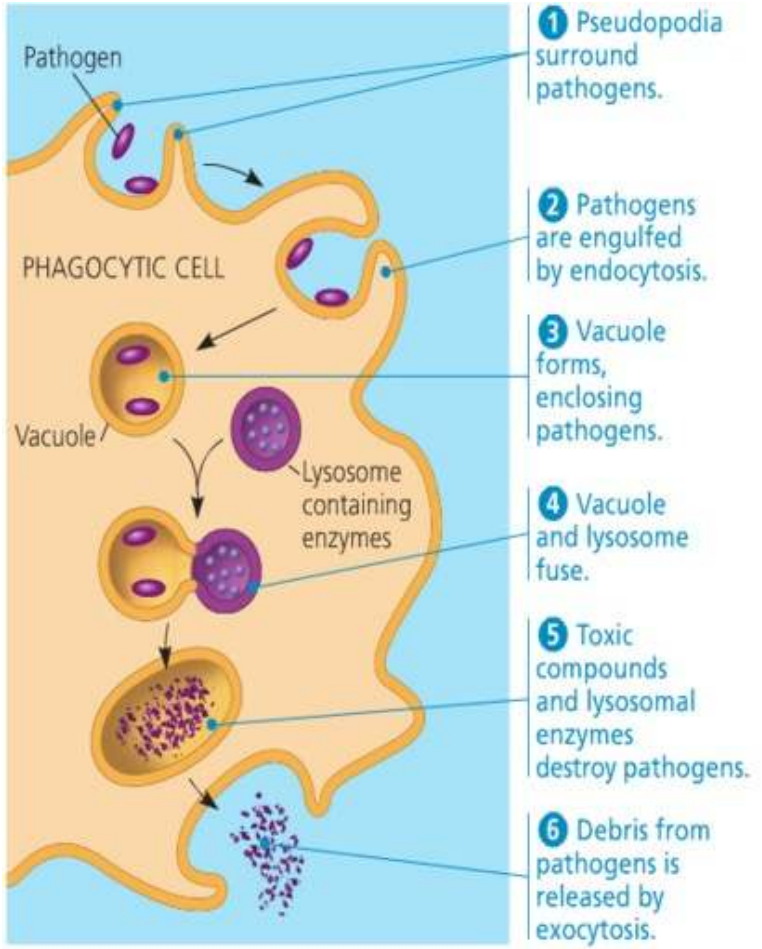
3.Killing and digestion

4.Oxygen-independent mechanisms: low pH, lysozyme, lactoferrin, proteolytic/hydrolytic enzymes

5.Oxygen dependent mechanisms



Phagocytosis



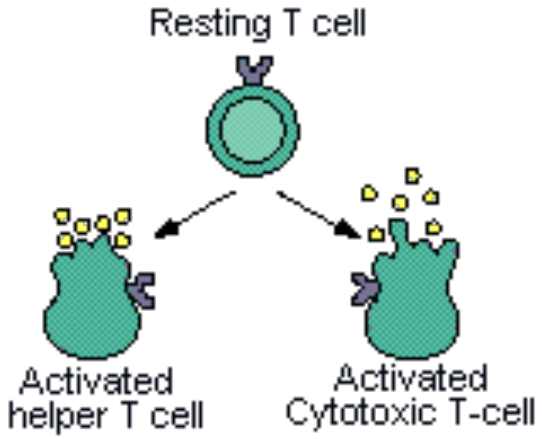
T cell

T Cells

T cells have two major roles in immune defense. Regulatory T cells are essential for orchestrating the response of an elaborate system of different types of immune cells.

Helper T cells, for example, also known as CD4 positive T cells (**CD4+ T cells**), alert B cells to start making antibodies; they also can activate other T cells and immune system scavenger cells called **macrophages** and influence which type of antibody is produced.

Certain T cells, called CD8 positive T cells (**CD8+ T cells**), can become killer cells that attack and destroy infected cells. The **killer T cells** are also called **cytotoxic T cells** or CTLs (cytotoxic lymphocytes).

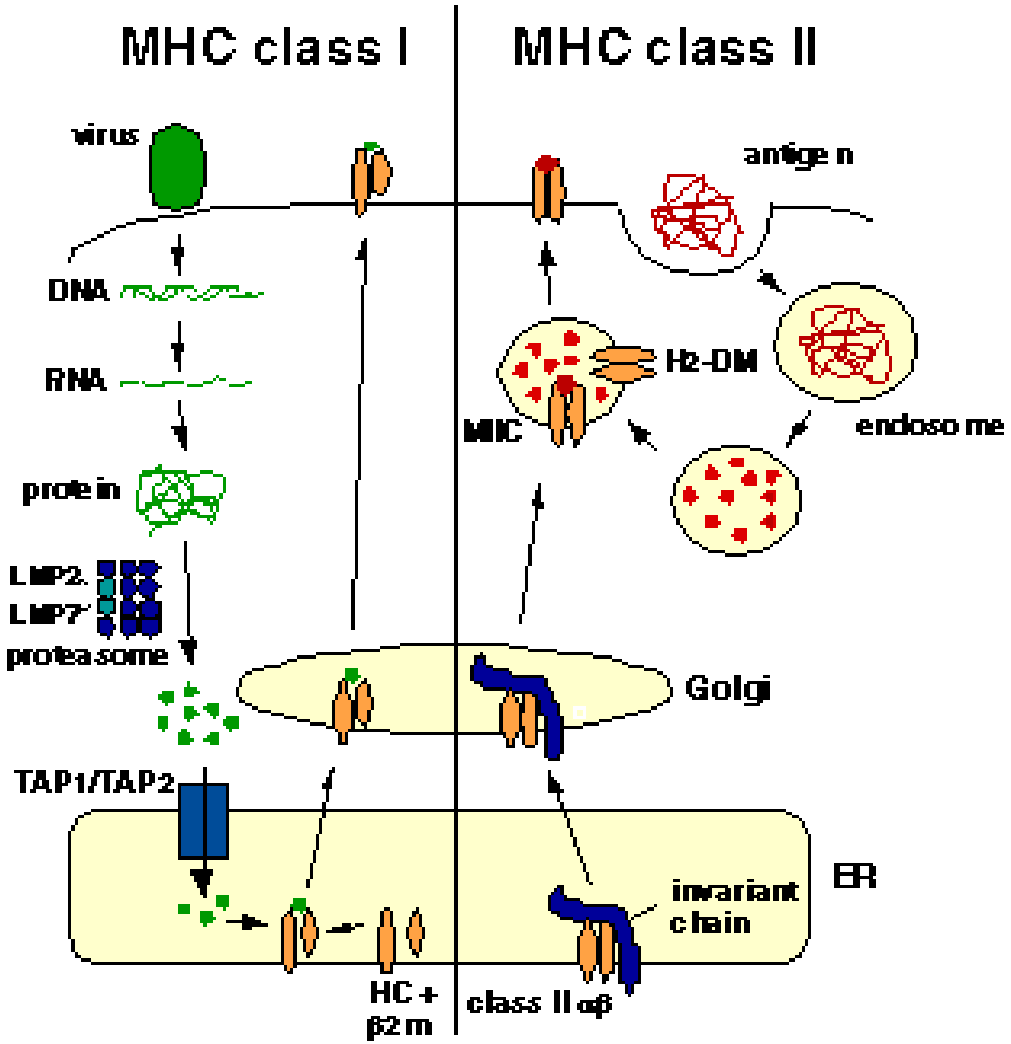




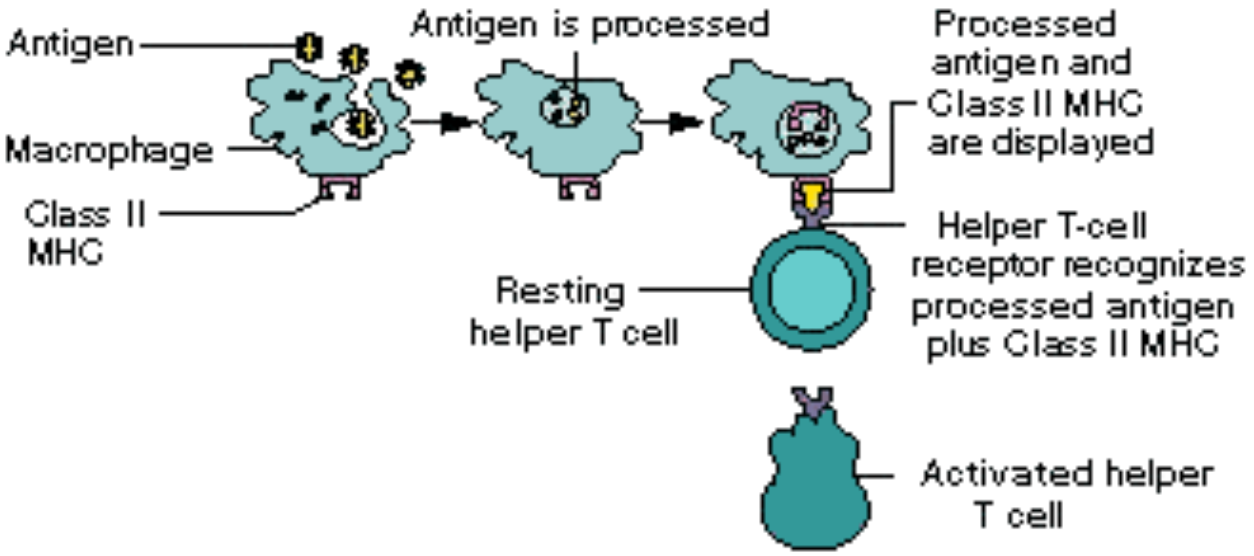




MHC

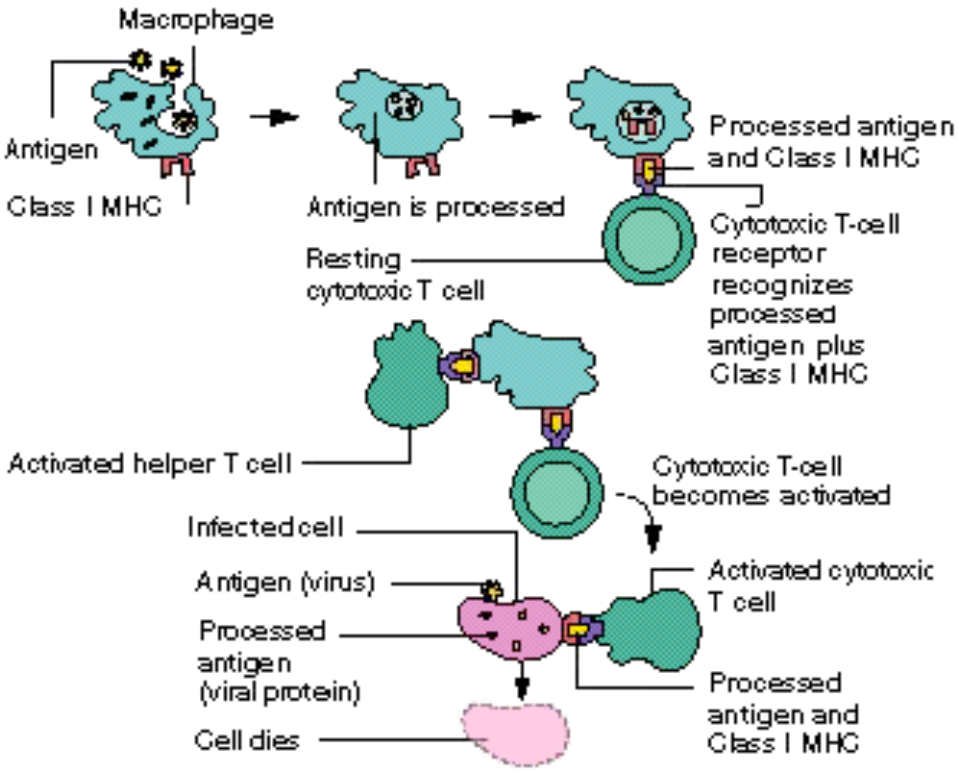

MHC class I
MHC class II
virus
antigen
DNA
RNA
protein
LMP2
LMP7
proteasome
H2-DM
MHC
endosome
Golgi
TAP1/TAP2
ER
invariant chain
HC + β2m
class II αβ

T cell-MHC-II



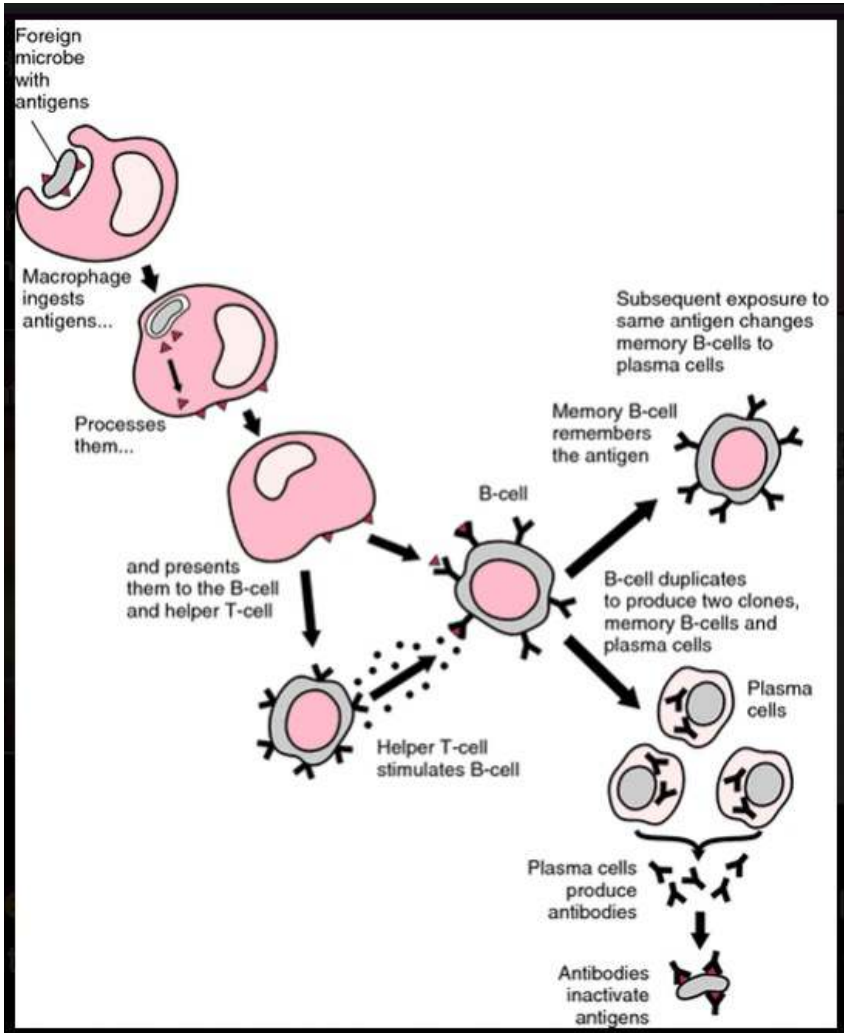
After it engulfs and processes an antigen, the macrophage displays the antigen fragments combined with a Class II MHC protein on the macrophage cell surface. The antigen-protein combination attracts a helper T cell, and promotes its activation.

MHC-I



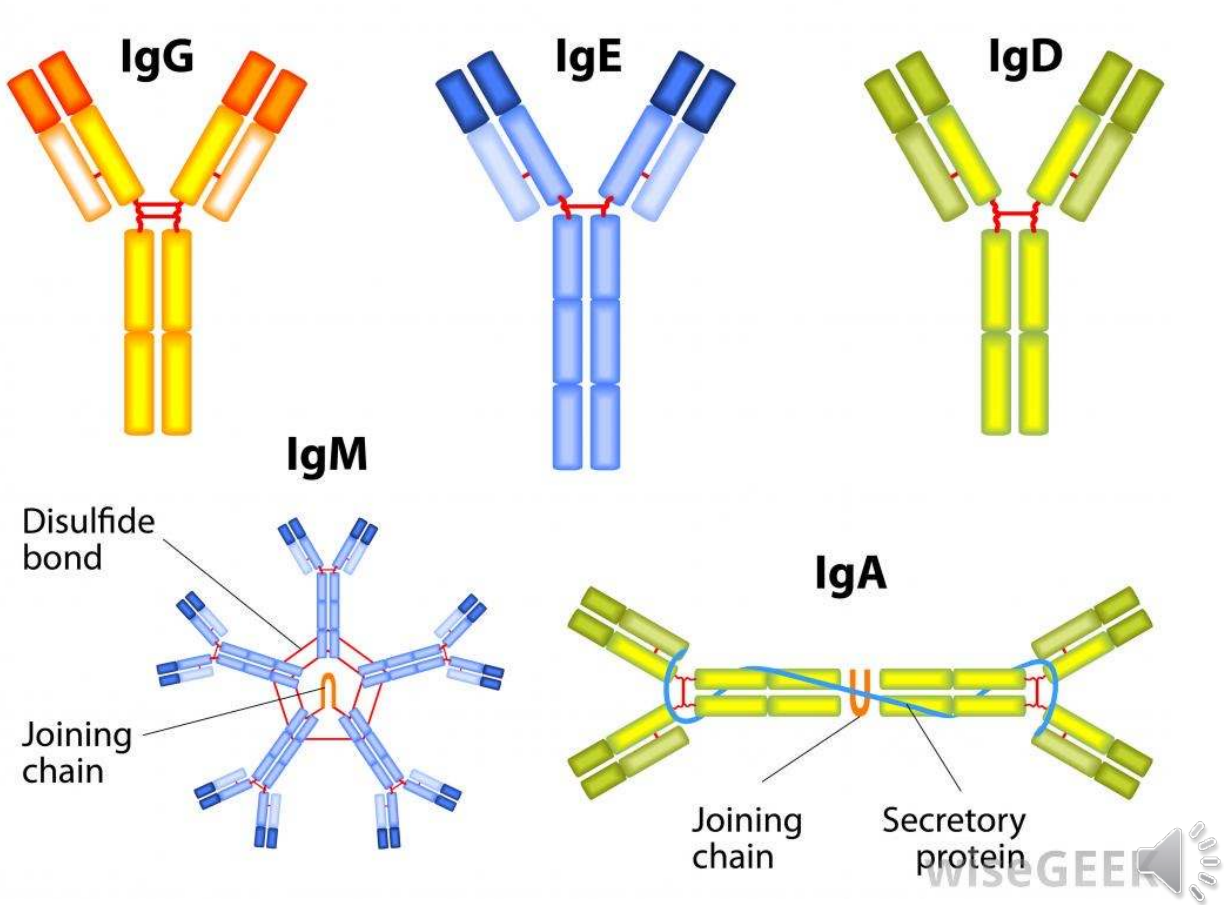
After a macrophage engulfs and processes an antigen, the macrophage displays the antigen fragments combined with a Class I MHC protein on the macrophage cell surface. A receptor on a circulating, resting cytotoxic T cell recognizes the antigen-protein complex and binds to it. The binding process and a helper T cell activate the cytotoxic T cell so that it can attack and destroy the diseased cell.

antibody

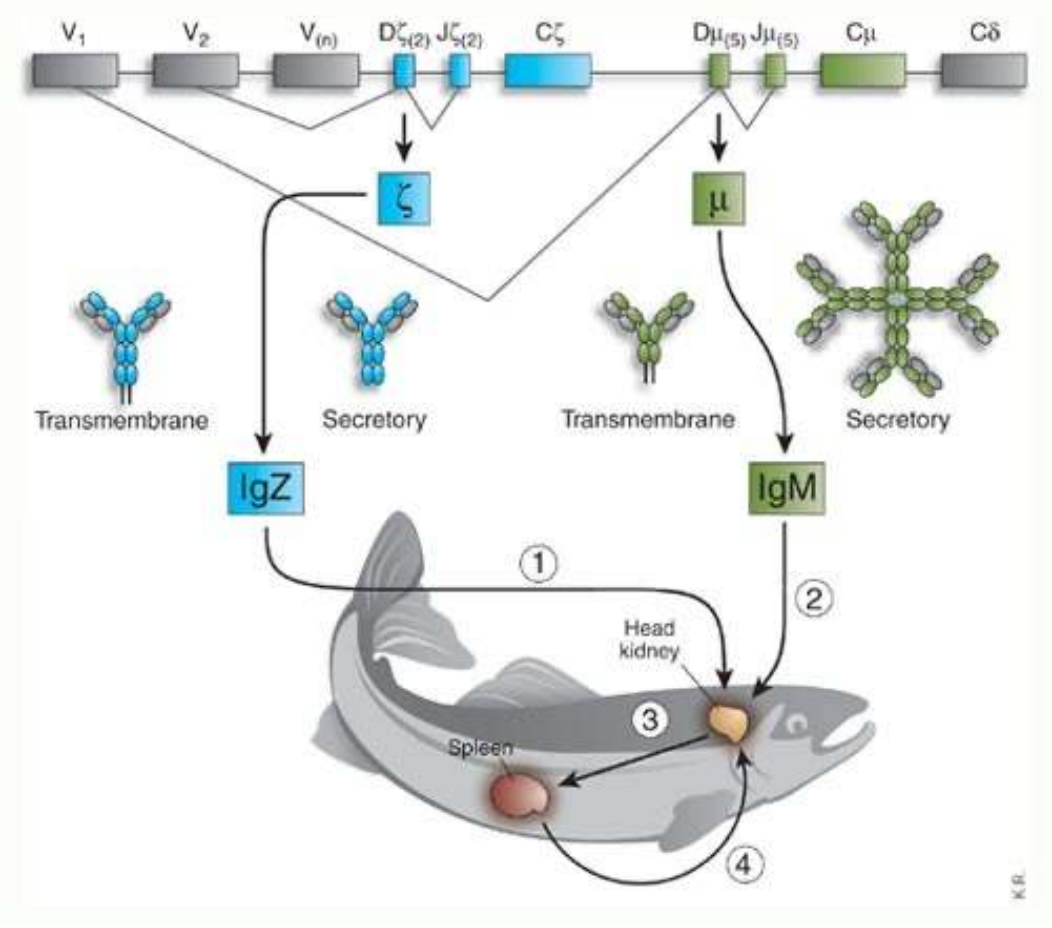

Foreign microbe with antigens
Macrophage ingests antigens...
Processes them...
and presents them to the B-cell and helper T-cell
Helper T-cell stimulates B-cell
B-cell
B-cell duplicates to produce two clones, memory B-cells and plasma cells
Memory B-cell remembers the antigen
Subsequent exposure to same antigen changes memory B-cells to plasma cells
Plasma cells
Plasma cells produce antibodies
Antibodies inactivate antigens



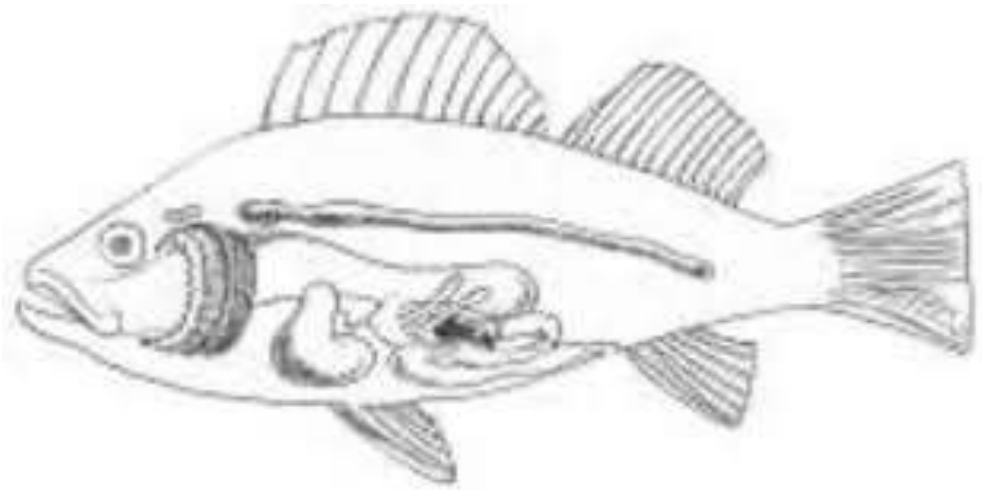
Antibody in human



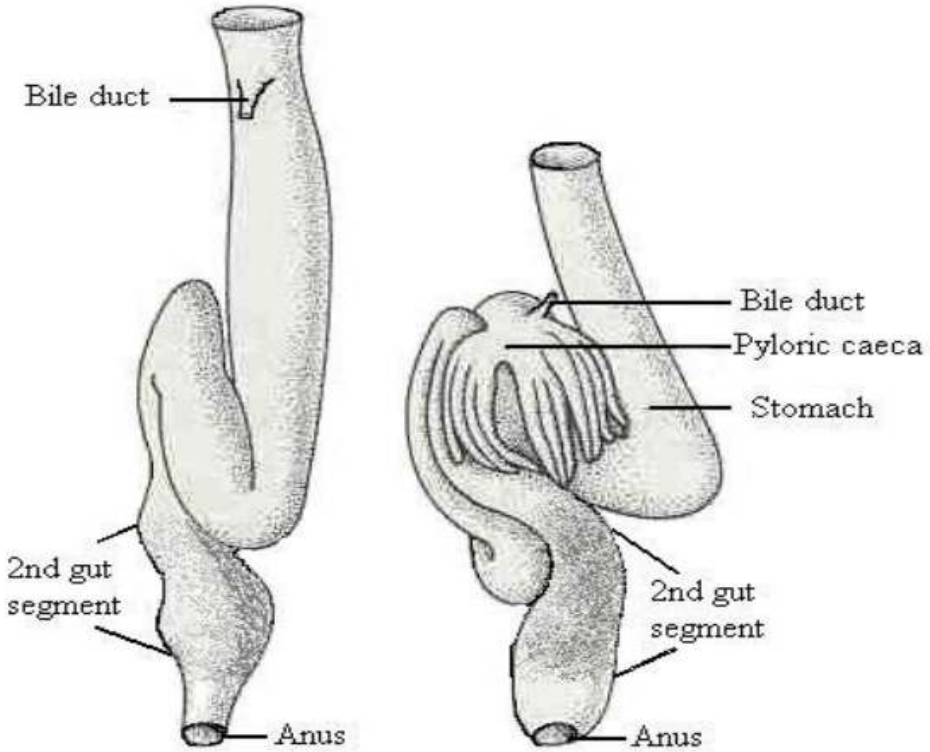
Antibody in fish



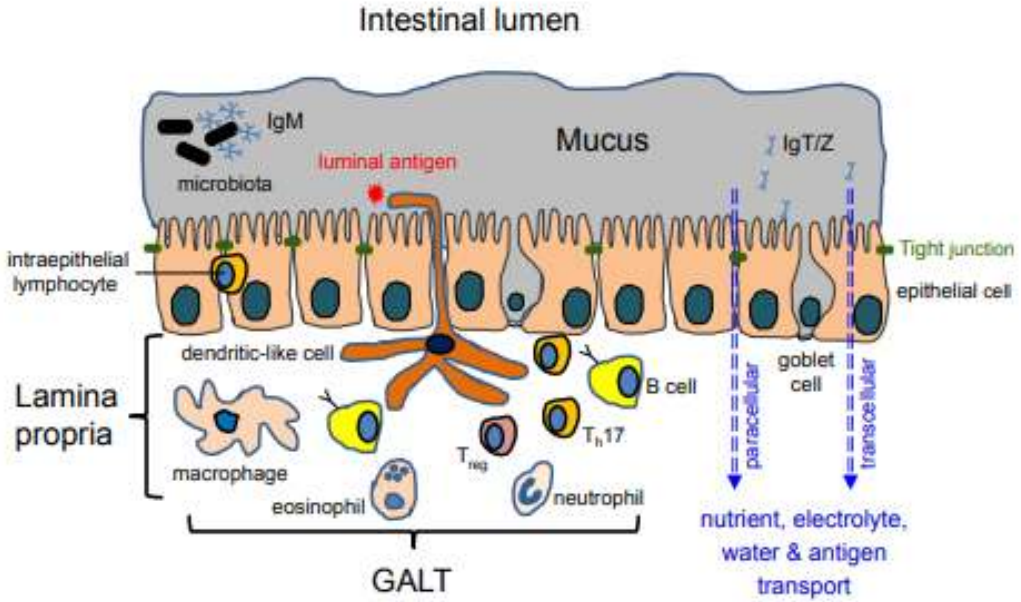
Intestinal immune response mechanism



Intestinal immune response mechanism



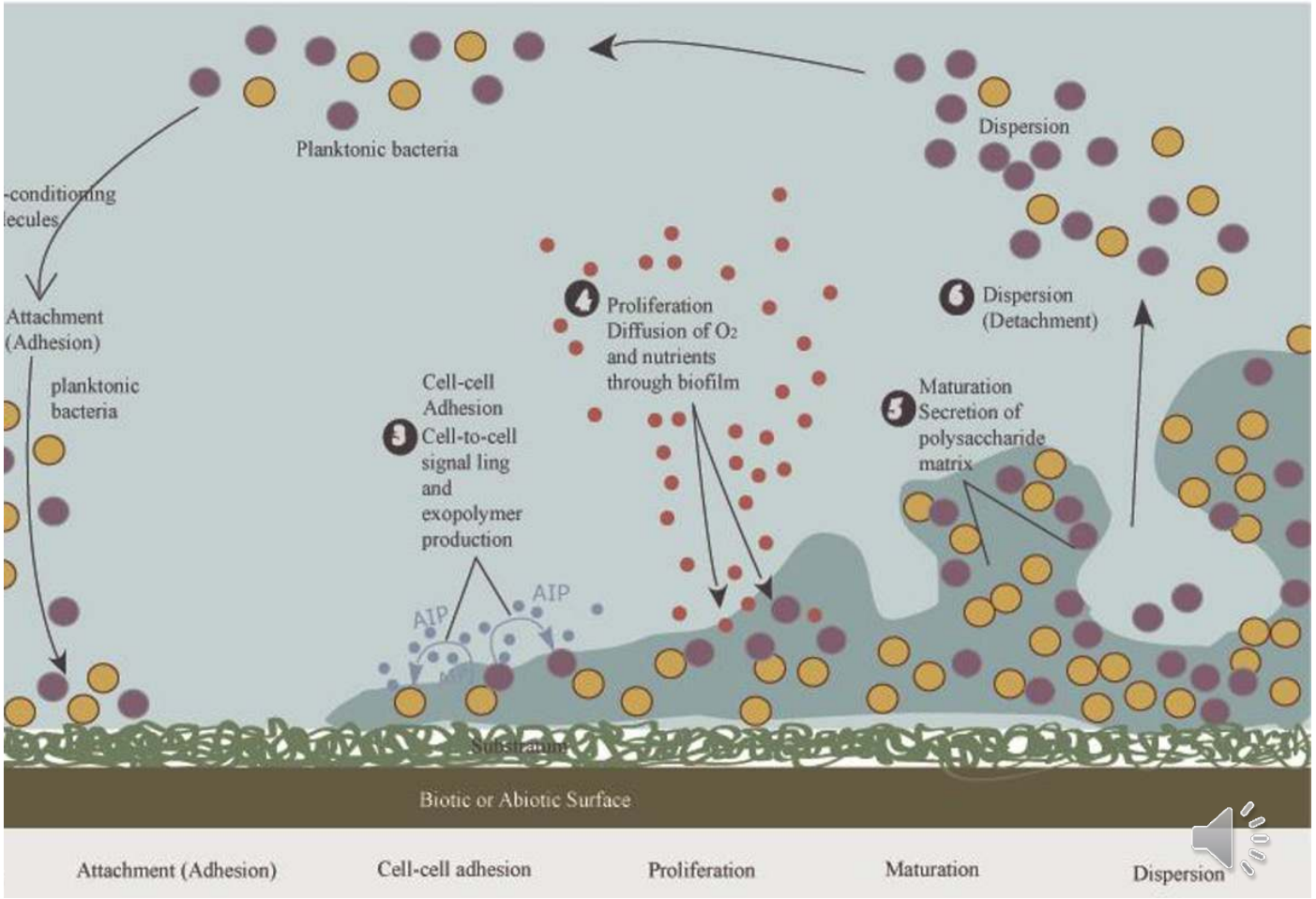
Intestinal immune response mechanism



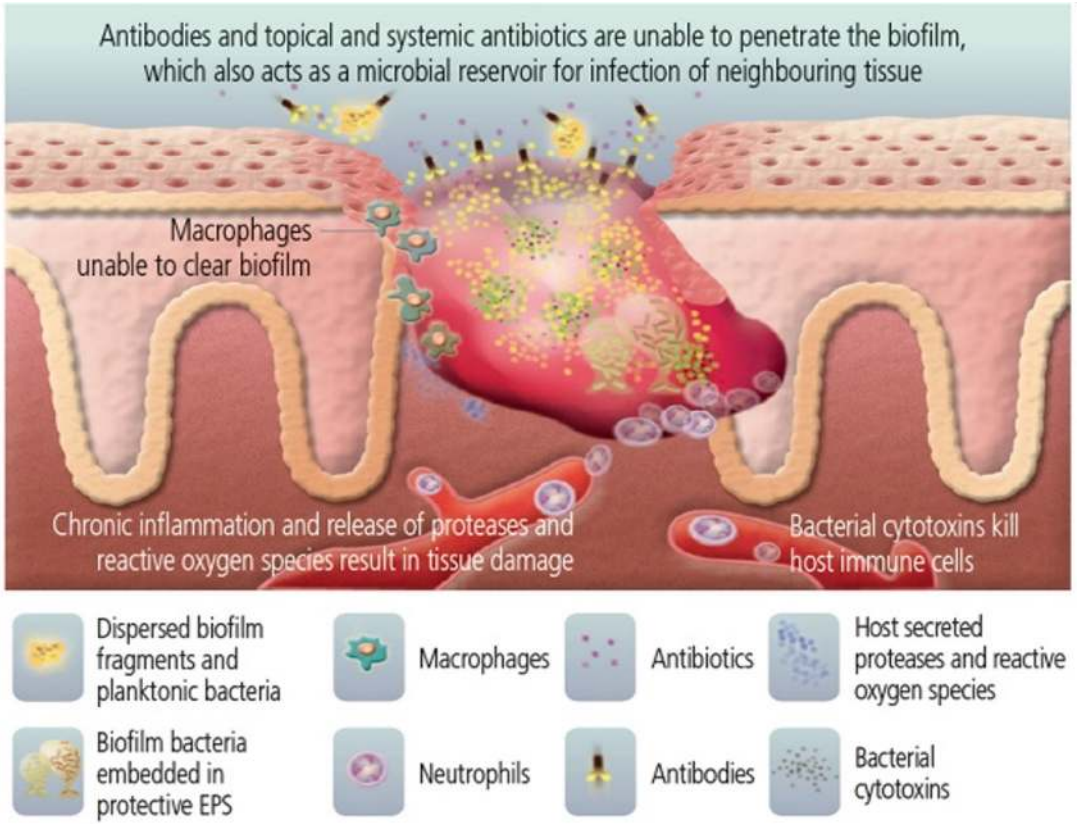
- 魚類免疫的現況



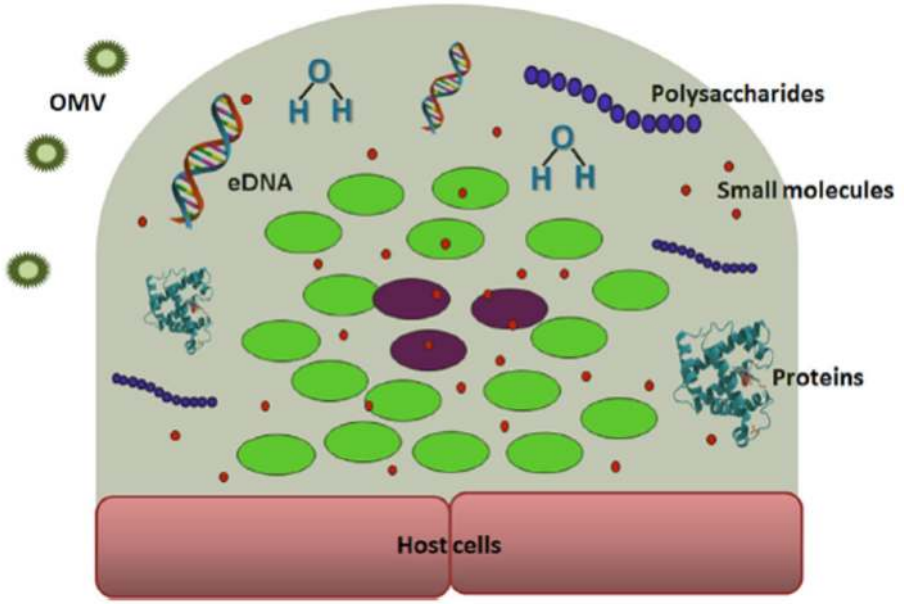
Biofilm


Planktonic bacteria
Dispersion
-conditioning
lecules
Attachment
(Adhesion)
planktonic
bacteria
4 Proliferation
Diffusion of O_2
and nutrients
through biofilm
6 Dispersion
(Detachment)
Cell-cell
Adhesion
3 Cell-to-cell
signal ling
and
exopolymer
production
5 Maturation
Secretion of
polysaccharide
matrix
AIP
AIP
Substratum
Biotic or Abiotic Surface
Attachment (Adhesion)
Cell-cell adhesion
Proliferation
Maturation
Dispersion

Biofilm


Antibodies and topical and systemic antibiotics are unable to penetrate the biofilm, which also acts as a microbial reservoir for infection of neighbouring tissue
Macrophages unable to clear biofilm
Chronic inflammation and release of proteases and reactive oxygen species result in tissue damage
Bacterial cytotoxins kill host immune cells
Dispersed biofilm fragments and planktonic bacteria
Macrophages
Antibiotics
Host secreted proteases and reactive oxygen species
Biofilm bacteria embedded in protective EPS
Neutrophils
Antibodies
Bacterial cytotoxins

Biofilm



(Watters, Fleming et al. 2016)

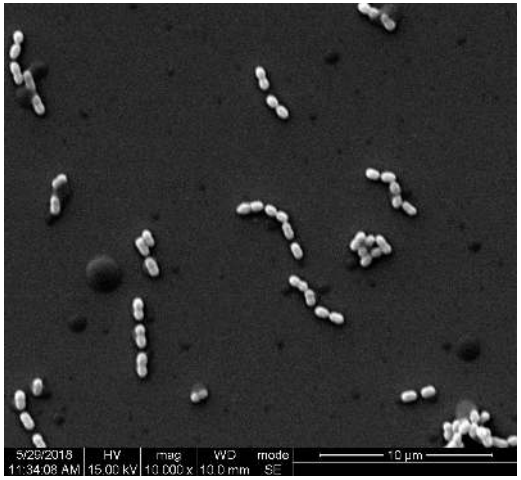
Biofilm

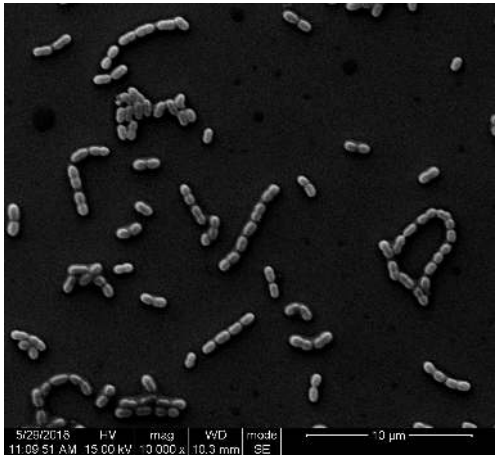
Erythromycin							
µg/mL	<1	1	64	256	1024	4096	16384
MIC	21	0	0	1	0	0	0
MBEC*	0	0	0	0	0	0	22
Florfenicol							
µg/mL	<1	1	64	256	1024	4096	16384
MIC	22	0	0	0	0	0	0
MBEC*	0	0	0	0	0	0	22



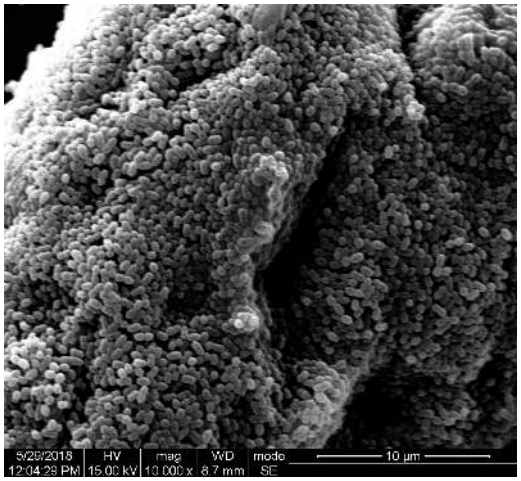
Biofilm

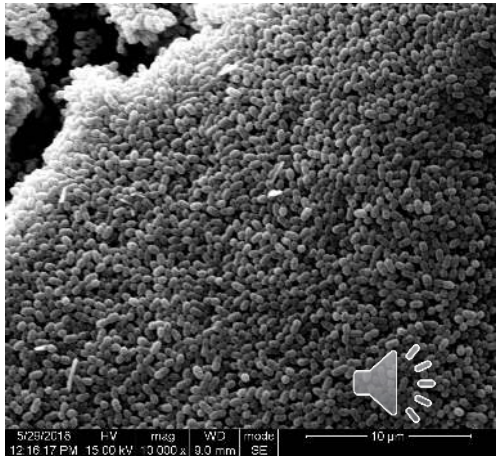
Planktonic





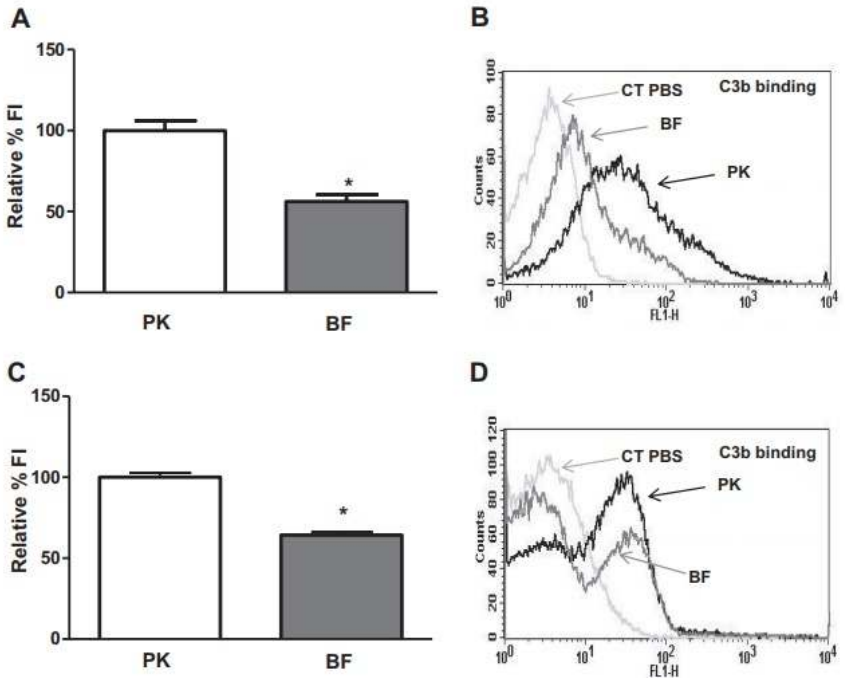
Biofilm





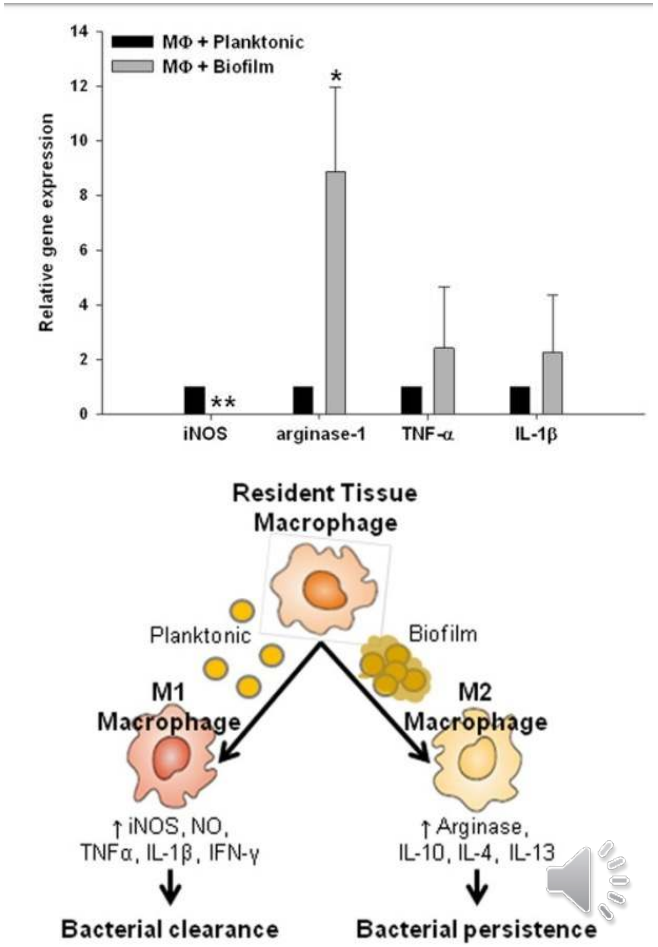
Bacteria in biofilm avoid complement immunity

- The present study confirms that biofilm formation in *Streptococcus. Pneumoniae* is an efficient means of evading both the classical and the pspc-dependent alternative complement pathways the host immune system.(Domenech, ramos-sevillano et al. 2013)



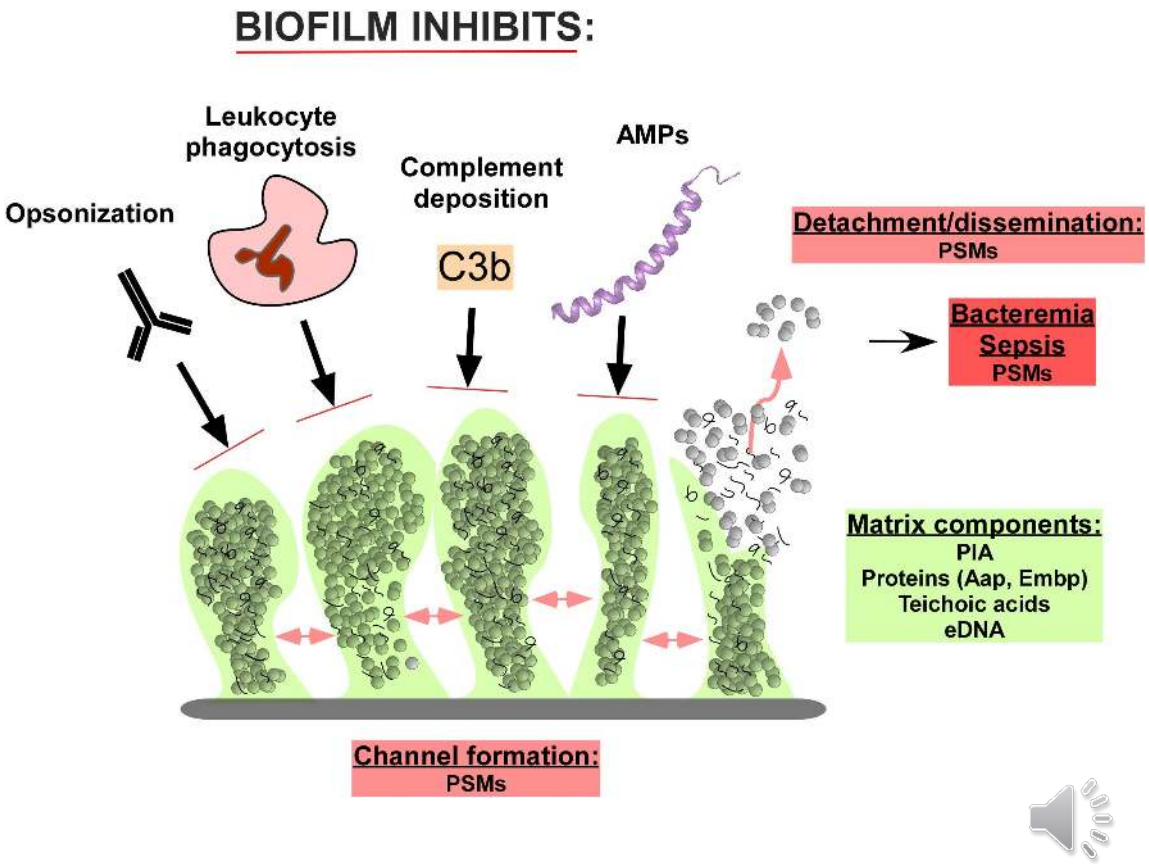
Biofilm infection change the macrophages from M0 to M2

- Extracellular proteins are important constituents of biofilms 。
- *S. Aureus* has more than 30 extracellular proteins, leading to different immune regulation, glycolysis, DNA metabolism, and stress response(brady, leid et al. 2006)
- *S. Aureus* biofilms skew macrophage gene expression toward an alternatively activated M2 phenotype(thurlow, hanke et al. 2011)



The interaction of biofilm against host immune system

Biofilm



組織潰爛



vaccine

- Vaccination is the most effective method of combating disease.(Plant & Lapatra, 2011)



injection



oral



immersion

injection

在硬骨魚科(teleost)注射抗原後之免疫反應是由脾臟之 melanomacrophage centers 及 橢球體(ellipsoids)產生。



oral

口服型疫苗就是直接將疫苗添加入餌料內施用。此型疫苗使用最為理想、方便，也可 減少魚之緊迫壓力，使用量較浸泡型疫苗為少，頗為經濟，實為最適用於稚魚免疫之方法。



oral

- 酒精包埋法
- 飼料混合重新朔型



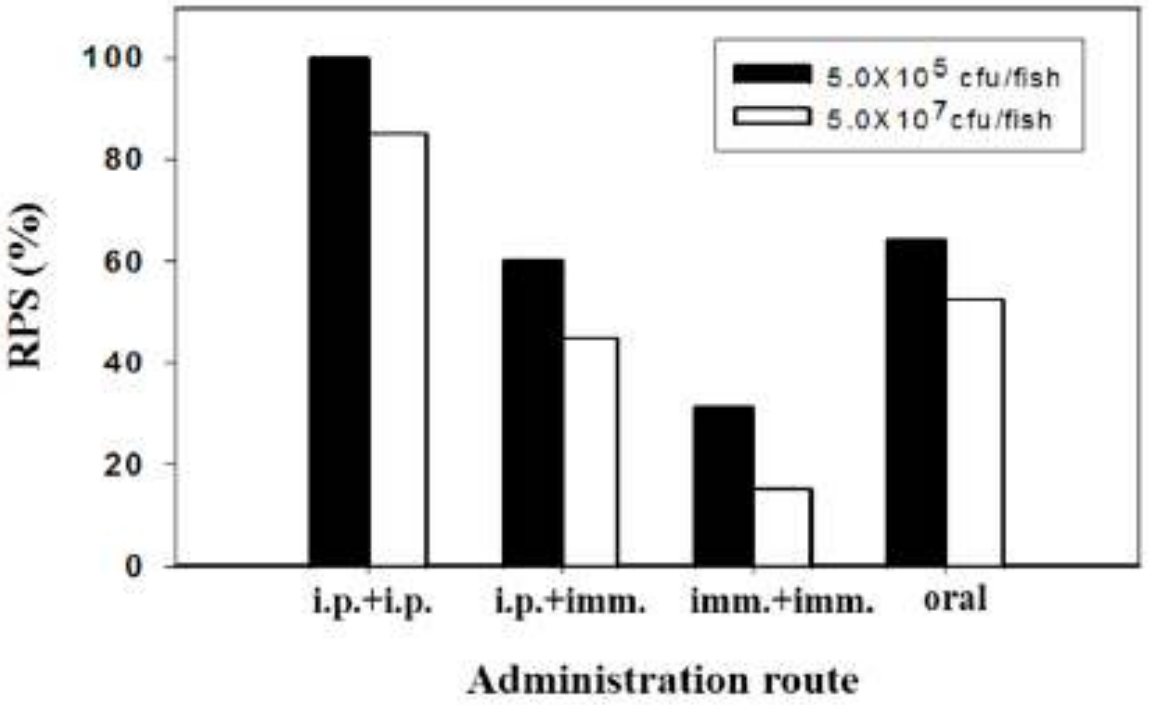
immersion

操作簡單

容易汙染

免疫機制尚待闡明

Vaccine-method



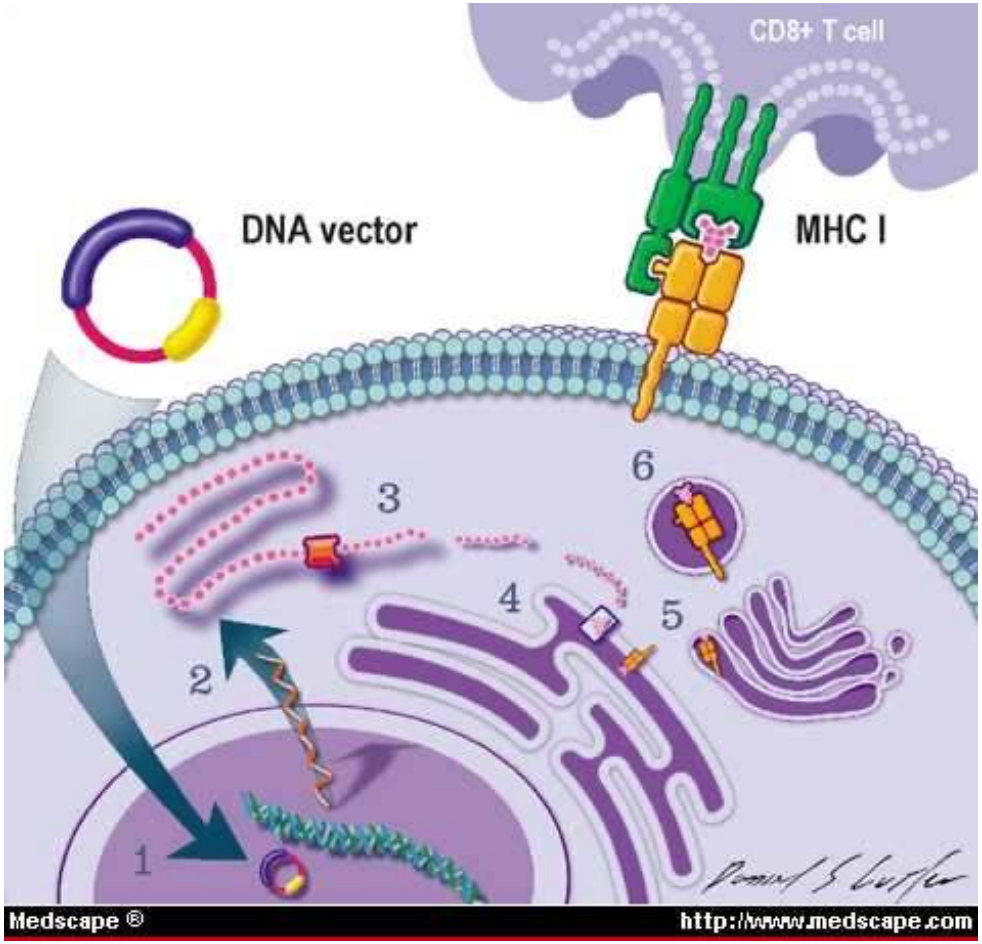
Vaccine

傳統疫苗

- DNA疫苗
- 減毒疫苗
- 次單位疫苗
- 不活化疫苗
- 類毒素疫苗



DNA vaccine


CD8+ T cell
DNA vector
MHC I
1
2
3
4
5
6
Medscape ®
http://www.medscape.com





DNA vaccine

- A number of theoretical safety concerns may be considered for DNA vaccines. These include:

– the fate of the plasmid in the vaccinated animals

– the risk of the integration of vaccine DNA sequences into the genome of the host, and subsequent negative side effects such as development of disease or integration into the germ line followed by vertical transfer

– the risk of inducing an anti-DNA immune response.



類毒素疫苗

- 類毒素疫苗(toxoid)是將具有毒性之細菌毒素，通常為外毒素(exotoxin)經過化學或物理方法使之失去毒性，而保留其免疫性，即為類毒素(toxoid)



次單位疫苗

- 首先從病原體基因體中選殖此抗原蛋白的基因，然後利用遺傳工程技術，將此基因送入其他細胞(如：大腸桿菌)內大量生產抗原蛋白質，接著收集、破壞大腸桿菌來進行抗原蛋白的純化，最後再製作成疫苗。




台視新聞
發現科學
10 17 18 20 28
"次單位疫苗"是什麼? 重組純化病毒蛋白
21:38:48

NNV

- 魚類免疫尚未完全感染病毒

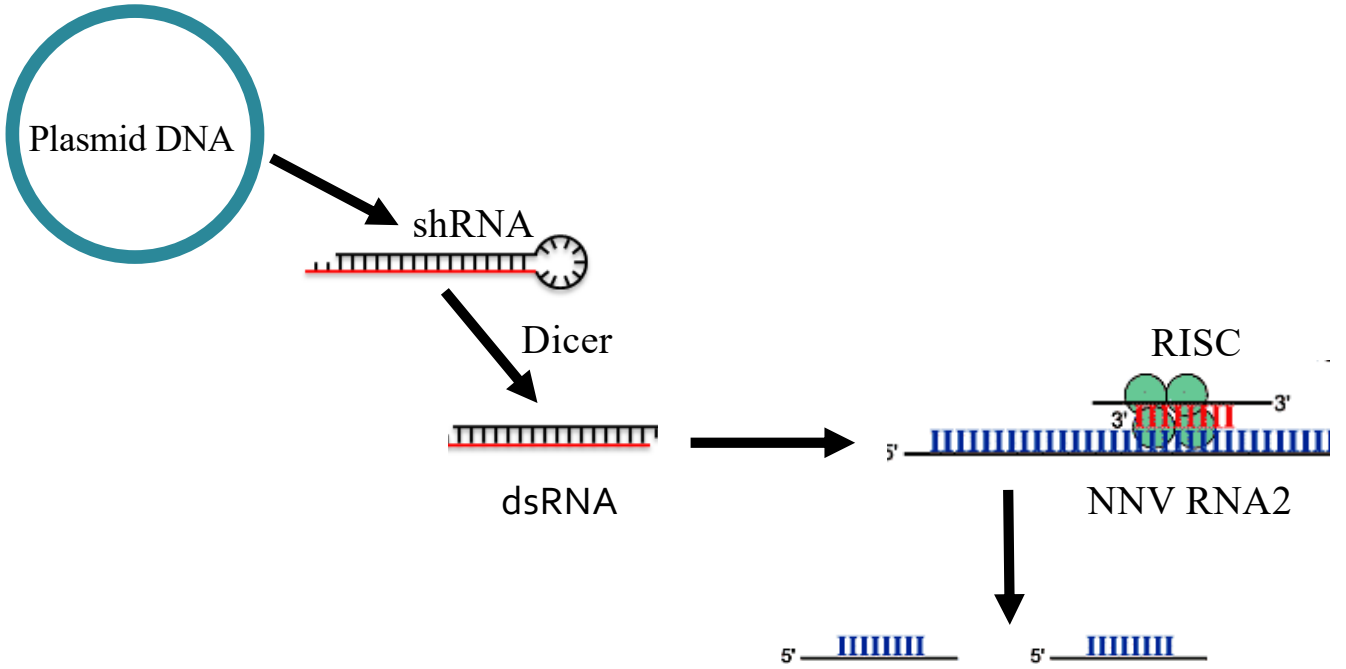


RNA Interference

- 在現今的研究中，siRNA可以有效且專一的抑制人類B型肝炎病毒基因的表現和病毒的複製。(Meng Z, Qiu S, Zhang X, Wu J, Schreiter T, et *al*. 2009)
- 在土撥鼠肝炎病毒(WHV)研究結果中顯示RNAi可以減少WHV的蛋白並且促進免疫基因的表現。 (Meng Z, Qiu S, Zhang X, Wu J, Schreiter T, et *al*. 2009)
- 在蝦類研究結果中說明dsRNA與siRNA可以減少黃頭病毒(YHV)與白點病毒(WSSV)基因的表現。(Shekhar and Lu 2009)



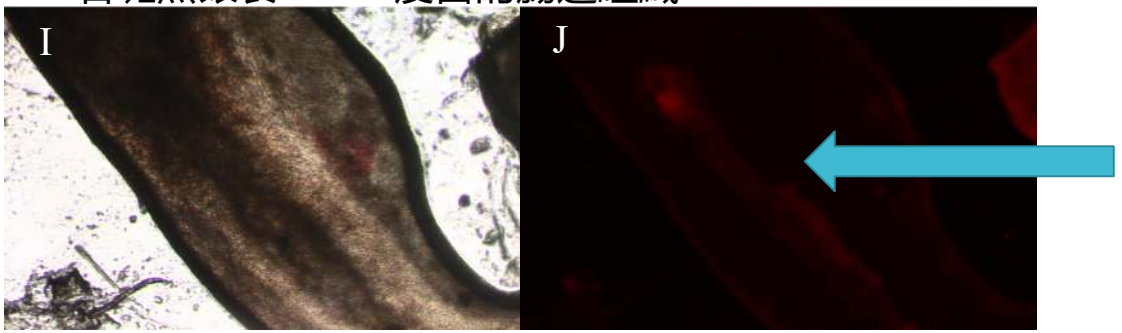
RNA Interference



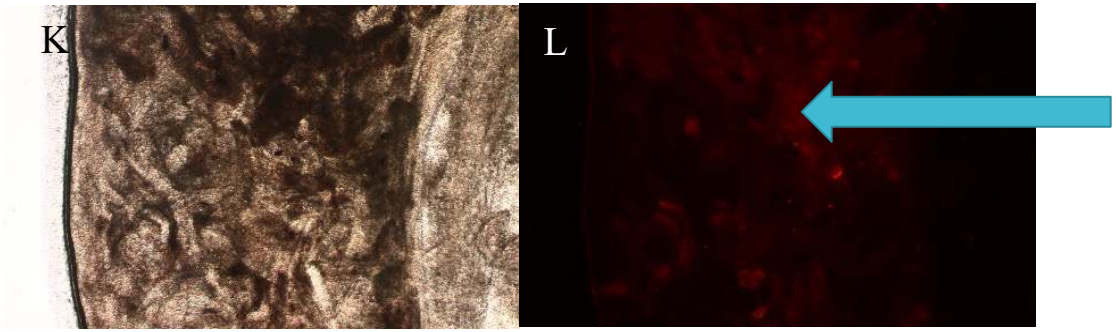
RNA Interference

明視野　　暗視野

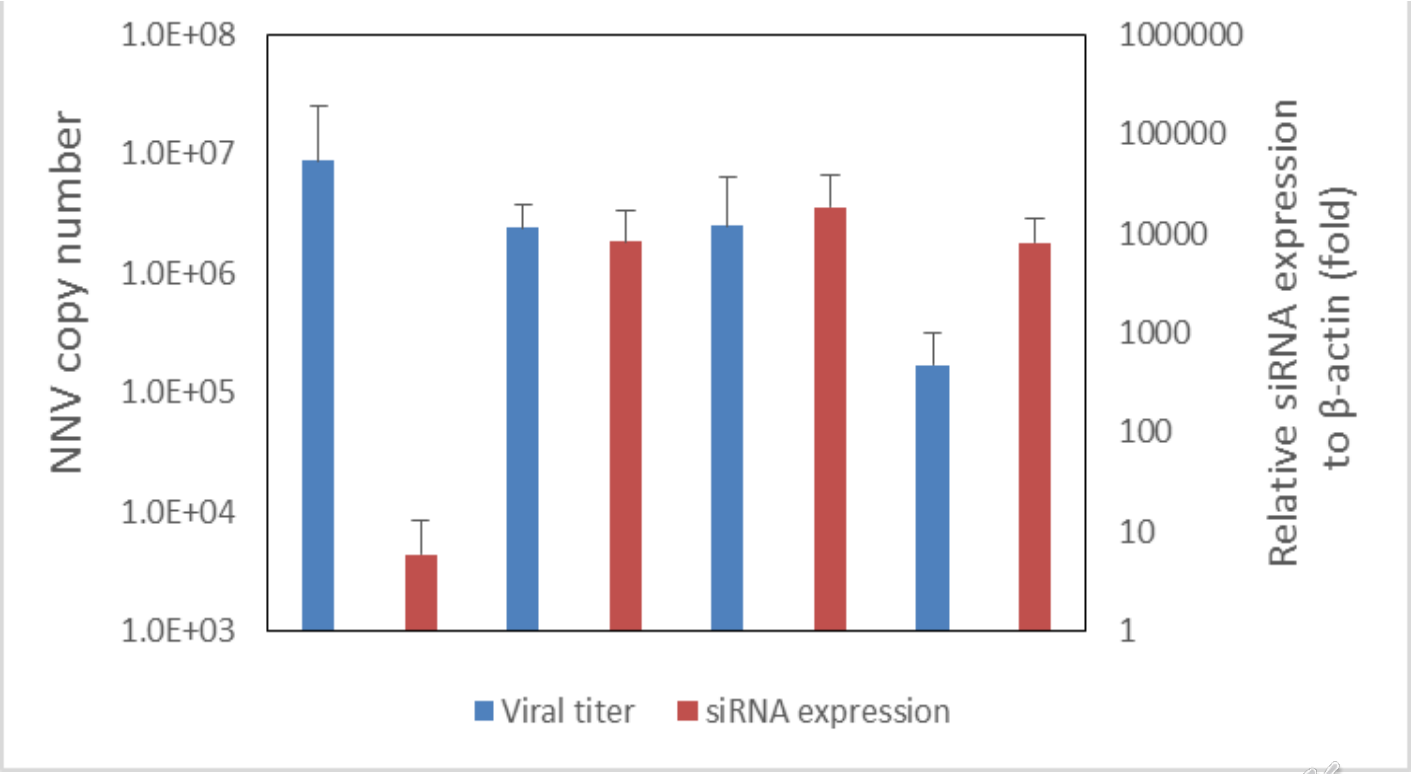
石斑魚餵食siRNA疫苗的腸道組織



石斑魚餵食siRNA疫苗的腸道組織



RNA Interference



不活化疫苗

- 不活化疫苗是指去活性的病毒、細菌整體或是微生物/寄生蟲的部分抗原。在 製造過程中通常以變性或化學方法修飾抗原決定位(epitopes)，去除微生物的活 性。一般在生產不活化疫苗時選擇與致病性相關的微生物作為材料，較能引起好 的免疫反應。



不活化疫苗

Table 1. Comparative efficacy of different vaccines against *Vibrio alginolyticus* in silver sea bream, *Sparus sarba*.

Vaccines	Number of Vaccinated Fish	Number of Challenged Fish	Number Of Dead Fish	Mortality (%)	RPS
Whole Cell Bacterins					
Formalin-killed	36	35	7	20	80
Phenol-killed	36	34	8	23.5	76.5
Chloroform-killed	36	35	25	71.4	28.6
Heat-killed	36	36	18	50	50
Subcellular Bacterins					
LPS	36	36	0	0	100
Control	36	36	36	100	0

LPS = lipopolysaccharide; RPS = relative percentage of survival.

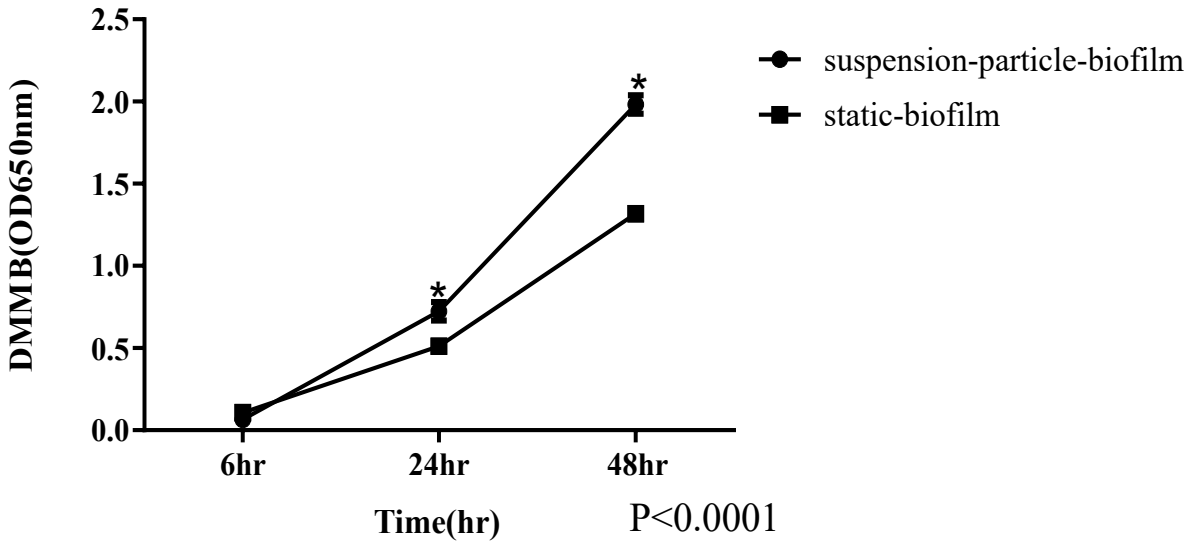


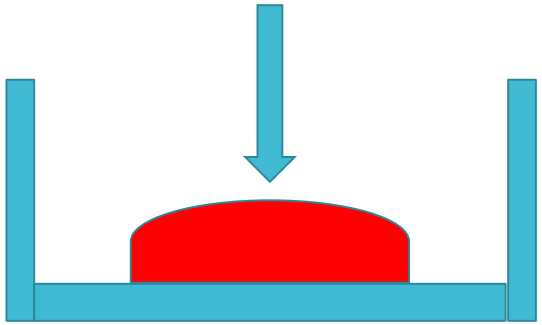
生物膜疫苗

- 有較多的免疫原性
- 製作繁瑣

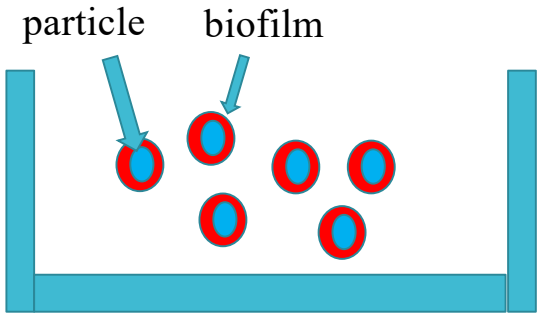


生物膜疫苗

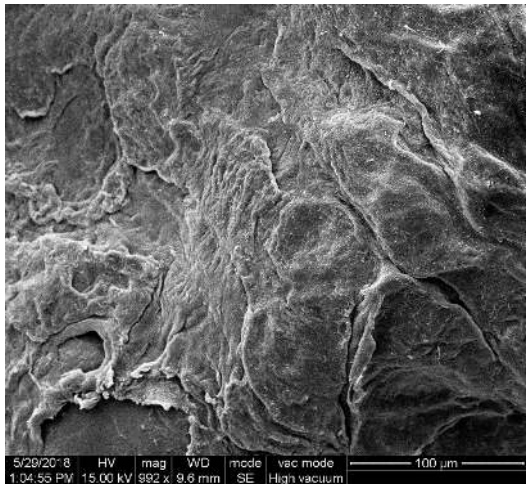


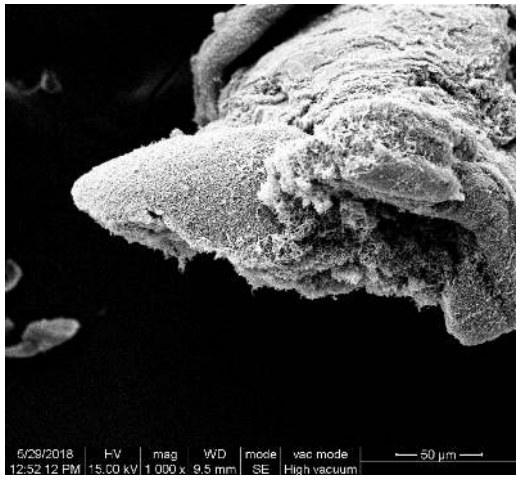


Static biofilm

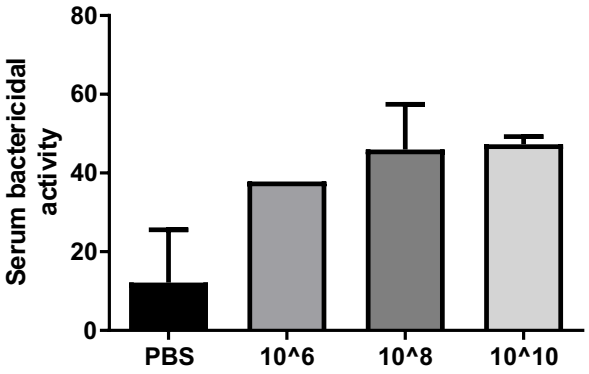


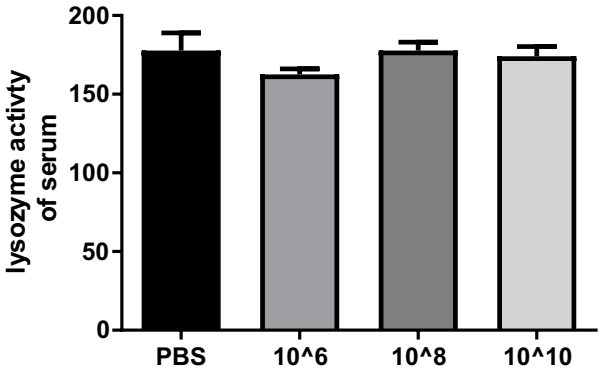
生物膜疫苗

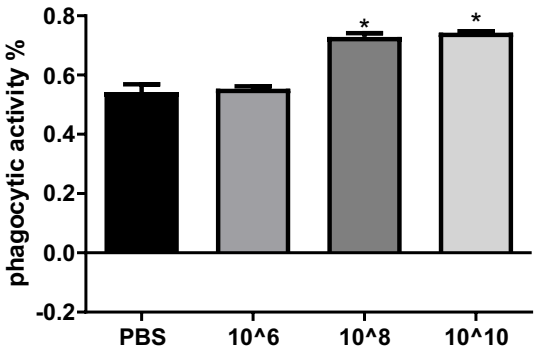

5/29/2018 HV mag WD mode vac mode
1:04:55 PM 15.00 kV 992 x 9.6 mm SE High vacuum
100 µm

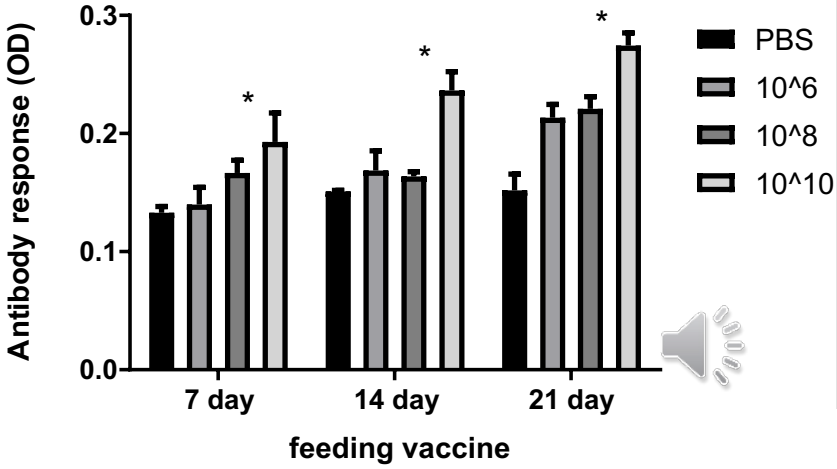

5/29/2018 HV mag WD mode vac mode
12:52:12 PM 15.00 kV 1 000 x 9.5 mm SE High vacuum
50 µm

生物膜疫苗

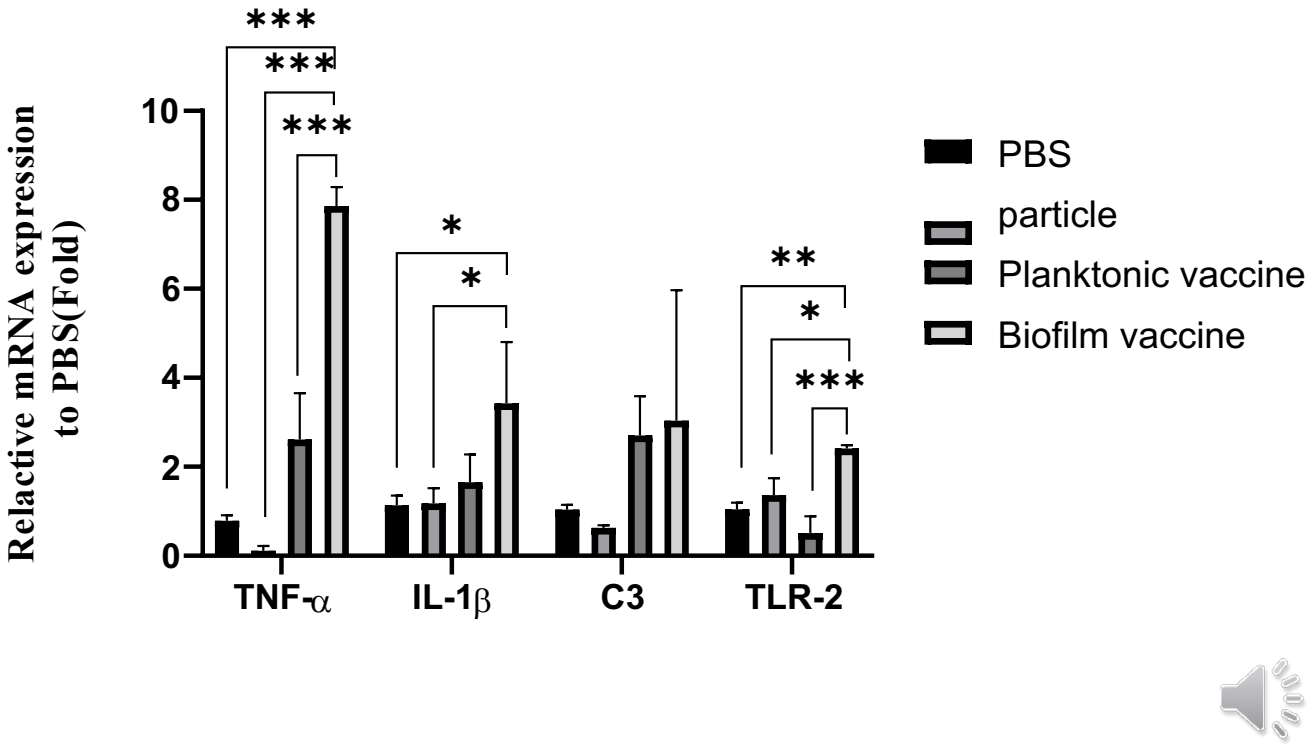

Serum bactericidal activity
80
60
40
20
0
PBS
10^6
10^8
10^10


lysozyme activty of serum
200
150
100
50
0
PBS
10^6
10^8
10^10

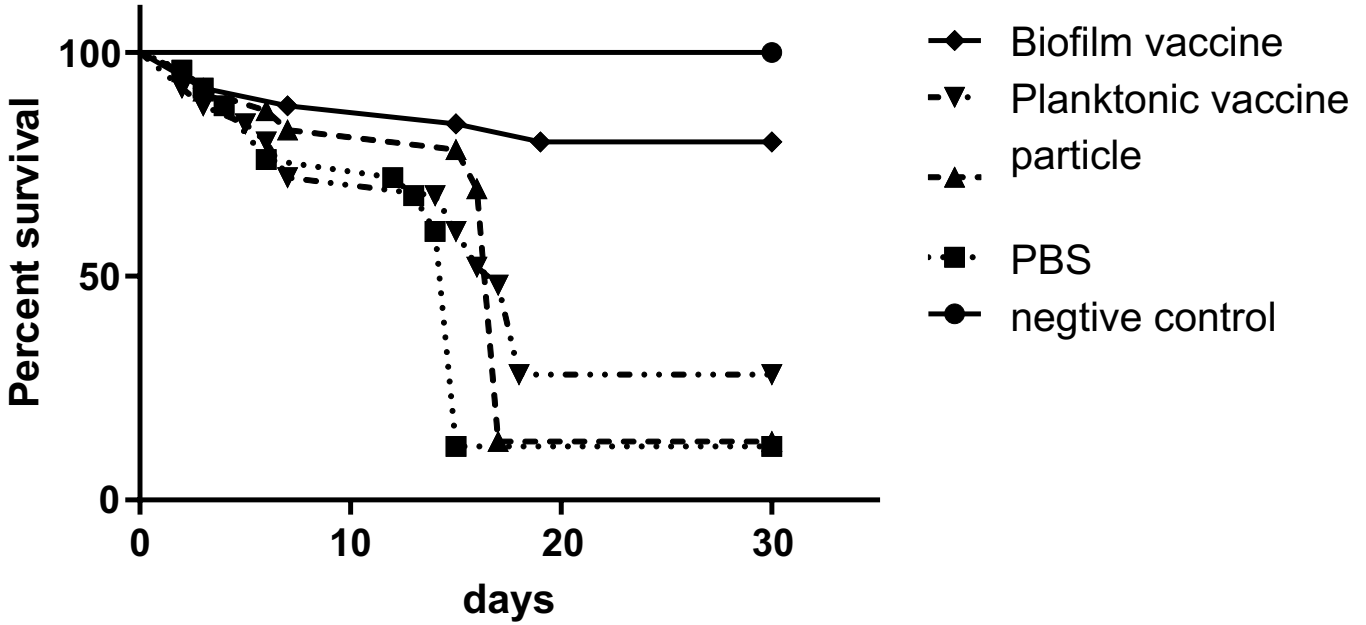

phagocytic activity %
0.8
0.6
0.4
0.2
0.0
-0.2
*
*
PBS
10^6
10^8
10^10


Antibody response (OD)
0.3
0.2
0.1
0.0
*
*
*
7 day
14 day
21 day
feeding vaccine
PBS
10^6
10^8
10^10

生物膜疫苗


Relactive mRNA expression to PBS(Fold)
10
8
6
4
2
0
***
***
***
*
*
**
*
***
TNF-α
IL-1β
C3
TLR-2
PBS
particle
Planktonic vaccine
Biofilm vaccine

生物膜疫苗


Percent survival
100
50
0
0
10
20
30
days
Biofilm vaccine
Planktonic vaccine
particle
PBS
negtive control

- 謝謝

